MECHANICS APPLIED TO AEROSPACE ENGINEERING - PART I

Topic 1: Fundamentals and Particle Kinematics - Problem Sets

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Ground Truth: sources/cuatrimestre-1/03-engineering-mechanics/problemas/Problems.pdf (2026 Edition)

Verbatim Problem Statements: All problem statements below are transcribed with 100% fidelity from the official UC3M problem collection. Solutions will be added in subsequent analytical development phases.

PROBLEM 01

Rocket Helical Trajectory

Original statement · Problems.pdf, p. 6

MIT OCW Benchmark

An out-of-control rocket traces an upward helical trajectory described in Cartesian coordinates for $t \geq 0$ by:

$$x = a\cos(\omega t) \tag{1}$$

$$y = a\sin(\omega t) \tag{2}$$

$$z = v_z t \tag{3}$$

where a, v_z, and ω are constants.

(a)

Express the position, velocity, and acceleration vectors in the Cartesian x, y, z system using unit vectors $\mathbf{i}, \mathbf{j}, \mathbf{k}$.

(b) Express the position, velocity, and acceleration vectors in the cylindrical coordinate system using unit vectors $\mathbf{e}_R, \mathbf{e}_\theta, \mathbf{e}_z$.

(c) Express the velocity and acceleration vectors using s, ρ and the vector basis $\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b$.

(d) Compute the modulus of the velocity vector.

(e) Express the evolution of the arc-length coordinate as a function of time.

(f) Write down the trajectory as a function of the arc-length coordinate.

(g) Where is the center of curvature for this curve when the particle is located at $(x, y, z) = (a, 0, 0)$?

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** An out-of-control sounding rocket ascends following a regular circular helix around the vertical z-axis.
- **Hypotheses:** The cylinder radius $a > 0$, vertical ascent velocity $v_z > 0$, and angular spin rate $\omega > 0$ are strictly positive constants. The rocket is modeled as an ideal point particle P.
- **Configuration Degrees of Freedom (CDOFs):** The trajectory is purely kinematically prescribed by time t; thus, the particle has $\mathrm{CDOF} = 1$ parameterized by time t or arc length s.

- **Parameters & SI Units:** Radius a [m], angular frequency ω [rad/s], vertical velocity v_z [m/s], time t [s].

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Inertial Reference Frame:** $\mathcal{S}_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$, fixed to the launch site with vertical axis $\mathbf{k}_0$ aligned with z.
- **Cylindrical Moving Basis:** $\mathcal{B}_{\text{cyl}} = \{\mathbf{e}_R, \mathbf{e}_\theta, \mathbf{e}_z\}$.

The transformation matrix from the inertial Cartesian basis $\mathcal{B}_0$ to the cylindrical basis $\mathcal{B}_{\text{cyl}}$ is the planar rotation matrix about $\mathbf{k}_0$:

$$[{}_0R_{\text{cyl}}] = \begin{pmatrix} \cos(\omega t) & -\sin(\omega t) & 0 \\ \sin(\omega t) & \cos(\omega t) & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

The basis vector relationships are given by matrix-vector multiplication:

$$\begin{pmatrix} \mathbf{e}_R \\ \mathbf{e}_\theta \\ \mathbf{e}_z \end{pmatrix} = [{}_0R_{\text{cyl}}]^T \begin{pmatrix} \mathbf{i}_0 \\ \mathbf{j}_0 \\ \mathbf{k}_0 \end{pmatrix} = \begin{pmatrix} \cos(\omega t)\mathbf{i}_0 + \sin(\omega t)\mathbf{j}_0 \\ -\sin(\omega t)\mathbf{i}_0 + \cos(\omega t)\mathbf{j}_0 \\ \mathbf{k}_0 \end{pmatrix}$$

- **Time Derivatives of Unit Vectors (Poisson's Theorem, Notes Sec. 3.1, Eqs. 3.3–3.5):**

Since $\mathcal{B}_{\text{cyl}}$ rotates at angular velocity $\boldsymbol{\omega} = \omega \mathbf{k}_0 = \omega \mathbf{e}_z$:

$$\frac{d\mathbf{e}_R}{dt} = \boldsymbol{\omega} \times \mathbf{e}_R = (\omega \mathbf{e}_z) \times \mathbf{e}_R = \omega \mathbf{e}_\theta$$

$$\frac{d\mathbf{e}_\theta}{dt} = \boldsymbol{\omega} \times \mathbf{e}_\theta = (\omega \mathbf{e}_z) \times \mathbf{e}_\theta = -\omega \mathbf{e}_R$$

$$\frac{d\mathbf{e}_z}{dt} = \mathbf{0}$$

- **Orthonormality of $[{}_0R_{\text{cyl}}]$:** $\det[{}_0R_{\text{cyl}}] = \cos^2(\omega t) + \sin^2(\omega t) = 1$ and $[{}_0R_{\text{cyl}}][{}_0R_{\text{cyl}}]^T = I$ (checked symbolically with sympy).
- **Frenet basis from the cylindrical basis.** The components of $\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b$ in $\mathcal{B}_{\text{cyl}}$ (obtained in Phase 3, part (c)) are the columns of the second change-of-basis matrix, with $V = \sqrt{a^2\omega^2 + v_z^2}$:

$$[{}_{\text{cyl}}R_{\text{tnb}}] = \begin{pmatrix} 0 & -1 & 0 \\ a\omega/V & 0 & -v_z/V \\ v_z/V & 0 & a\omega/V \end{pmatrix}, \qquad \det = \frac{a^2\omega^2 + v_z^2}{V^2} = 1, \qquad RR^T = I$$

- The three columns have unit norm and are mutually orthogonal, so $RR^T = I$; the composition $[{}_0R_{\text{tnb}}] = [{}_0R_{\text{cyl}}][{}_{\text{cyl}}R_{\text{tnb}}]$ is again a proper rotation (determinant $1 \cdot 1 = 1$). Both determinants and the products RR^T were verified symbolically with sympy.

PHASE 3

Step-by-Step Mathematical Deduction

(a) Cartesian Position, Velocity, and Acceleration

Pedagogical Justification: The position vector $\mathbf{r}(t)$ is directly defined by its Cartesian coordinates $(x(t), y(t), z(t))$ relative to the fixed origin O. Velocity and acceleration are obtained by direct term-by-term differentiation in the fixed inertial basis $\mathcal{B}_0$, where $\frac{d\mathbf{i}_0}{dt} = \frac{d\mathbf{j}_0}{dt} = \frac{d\mathbf{k}_0}{dt} = \mathbf{0}$.

- **Position Vector:**

$$\mathbf{r}(t) = a\cos(\omega t)\mathbf{i}_0 + a\sin(\omega t)\mathbf{j}_0 + v_z t\mathbf{k}_0$$

- **Velocity Vector (Explicit Chain Rule):**

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt} = \frac{d}{dt}[a\cos(\omega t)]\mathbf{i}_0 + \frac{d}{dt}[a\sin(\omega t)]\mathbf{j}_0 + \frac{d}{dt}[v_z t]\mathbf{k}_0$$

Applying the chain rule $\frac{d}{dt}\cos(\omega t) = -\omega\sin(\omega t)$ and $\frac{d}{dt}\sin(\omega t) = \omega\cos(\omega t)$:

$$\mathbf{v}(t) = -a\omega\sin(\omega t)\mathbf{i}_0 + a\omega\cos(\omega t)\mathbf{j}_0 + v_z\mathbf{k}_0$$

- **Acceleration Vector (Explicit Chain Rule):**

$$\mathbf{a}(t) = \frac{d\mathbf{v}}{dt} = \frac{d}{dt}[-a\omega\sin(\omega t)]\mathbf{i}_0 + \frac{d}{dt}[a\omega\cos(\omega t)]\mathbf{j}_0 + \frac{d}{dt}[v_z]\mathbf{k}_0$$

$$\mathbf{a}(t) = -a\omega^2\cos(\omega t)\mathbf{i}_0 - a\omega^2\sin(\omega t)\mathbf{j}_0 + 0\mathbf{k}_0 = -a\omega^2(\cos(\omega t)\mathbf{i}_0 + \sin(\omega t)\mathbf{j}_0)$$

(b) Cylindrical Position, Velocity, and Acceleration

> **Pedagogical Justification:** Cylindrical coordinates (R, θ, z) exploit the axial symmetry of the cylinder upon which the helix lies. Here, the radial distance $R(t) = \sqrt{x^2 + y^2} = a$ is constant, and the azimuthal angle is $\theta(t) = \omega t$.

- **Position in Cylindrical Coordinates:**

$$\mathbf{r}(t) = a\mathbf{e}_R + v_z t\mathbf{e}_z$$

- **Velocity Vector (Explicit Product Rule with Poisson Derivative):**

$$\mathbf{v}(t) = \frac{d}{dt}(a\mathbf{e}_R + v_z t\mathbf{e}_z) = a\frac{d\mathbf{e}_R}{dt} + \frac{d(v_z t)}{dt}\mathbf{e}_z + v_z t\frac{d\mathbf{e}_z}{dt}$$

Substituting $\frac{d\mathbf{e}_R}{dt} = \omega \mathbf{e}_\theta$ and $\frac{d\mathbf{e}_z}{dt} = \mathbf{0}$:

$$\mathbf{v}(t) = a\omega \mathbf{e}_\theta + v_z \mathbf{e}_z$$

- **Acceleration Vector (Explicit Differentiation):**

$$\mathbf{a}(t) = \frac{d}{dt}(a\omega \mathbf{e}_\theta + v_z \mathbf{e}_z) = a\omega \frac{d\mathbf{e}_\theta}{dt} + 0 = a\omega(-\omega \mathbf{e}_R) = -a\omega^2 \mathbf{e}_R$$

(c) Intrinsic Frenet-Serret Basis $\{\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b\}$

> **Pedagogical Justification:** The Frenet-Serret intrinsic basis (Notes Sec. 2.4) decomposes motion naturally along the curve itself. The tangent vector $\mathbf{e}_t$ points in the direction of instantaneous velocity, the principal normal $\mathbf{e}_n$ points toward the center of curvature, and the binormal $\mathbf{e}_b = \mathbf{e}_t \times \mathbf{e}_n$ completes the right-handed orthonormal triad.

- **Unit Tangent Vector $\mathbf{e}_t$:**

$$\mathbf{e}_t = \frac{\mathbf{v}}{\|\mathbf{v}\|} = \frac{a\omega \mathbf{e}_\theta + v_z \mathbf{e}_z}{\sqrt{a^2\omega^2 + v_z^2}}$$

- **Unit Principal Normal Vector $\mathbf{e}_n$:**

By definition of the intrinsic acceleration (Notes Eq. 2.17):

$$\mathbf{a} = \ddot{s}\mathbf{e}_t + \frac{\dot{s}^2}{\rho}\mathbf{e}_n$$

Since the speed $v = \dot{s} = \sqrt{a^2\omega^2 + v_z^2}$ is strictly constant in time, the tangential acceleration vanishes:

$$\ddot{s} = \frac{dv}{dt} = 0$$

Therefore, the entire acceleration is purely normal:

$$\mathbf{a} = a_n \mathbf{e}_n = -a\omega^2 \mathbf{e}_R$$

Taking the norm of both sides yields the magnitude of normal acceleration:

$$a_n = \|\mathbf{a}\| = a\omega^2$$

Dividing **a** by its norm gives the unit normal vector:

$$\mathbf{e}_n = \frac{\mathbf{a}}{a_n} = \frac{-a\omega^2 \mathbf{e}_R}{a\omega^2} = -\mathbf{e}_R$$

The principal normal points radially inward toward the central axis of the cylinder!

- **Radius of Curvature ρ:**

Using the relation $a_n = \frac{v^2}{\rho}$:

$$\rho = \frac{v^2}{a_n} = \frac{a^2\omega^2 + v_z^2}{a\omega^2} = a + \frac{v_z^2}{a\omega^2}$$

- **Unit Binormal Vector $\mathbf{e}_b$:**

Computing the vector cross product in the cylindrical basis:

$$\mathbf{e}_b = \mathbf{e}_t \times \mathbf{e}_n = \left(\frac{a\omega \mathbf{e}_\theta + v_z \mathbf{e}_z}{\sqrt{a^2\omega^2 + v_z^2}} \right) \times (-\mathbf{e}_R) = \frac{-a\omega(\mathbf{e}_\theta \times \mathbf{e}_R) - v_z(\mathbf{e}_z \times \mathbf{e}_R)}{\sqrt{a^2\omega^2 + v_z^2}}$$

Recalling cyclic cross products: $\mathbf{e}_\theta \times \mathbf{e}_R = -\mathbf{e}_z$ and $\mathbf{e}_z \times \mathbf{e}_R = \mathbf{e}_\theta$:

$$\mathbf{e}_b = \frac{-a\omega(-\mathbf{e}_z) - v_z\mathbf{e}_\theta}{\sqrt{a^2\omega^2 + v_z^2}} = \frac{-v_z\mathbf{e}_\theta + a\omega\mathbf{e}_z}{\sqrt{a^2\omega^2 + v_z^2}}$$

(d) Modulus of Velocity

> **Pedagogical Justification:** The speed $v(t) = \|\mathbf{v}(t)\|$ represents the scalar rate of distance traveled along the path.

- **Calculation:**

$$v = \|\mathbf{v}\| = \sqrt{(-a\omega\sin\omega t)^2 + (a\omega\cos\omega t)^2 + v_z^2} = \sqrt{a^2\omega^2(\sin^2\omega t + \cos^2\omega t) + v_z^2} = \sqrt{a^2\omega^2 + v_z^2}$$

The speed is constant for all $t \geq 0$.

(e) Evolution of Arc Length Coordinate $s(t)$

> **Pedagogical Justification:** The arc-length coordinate $s(t)$ measures the distance traversed along the trajectory from the initial position at $t = 0$. By fundamental calculus:

$$s(t) = \int_0^t v(t')\,dt'$$

- **Definite Integration:**

$$s(t) = \int_0^t \sqrt{a^2\omega^2 + v_z^2}\,dt' = \sqrt{a^2\omega^2 + v_z^2}\,[t']_0^t = t\sqrt{a^2\omega^2 + v_z^2}$$

(f) Trajectory as a Function of Arc Length s

> **Pedagogical Justification:** To express the curve intrinsically independent of time, we invert $s(t)$ to obtain time as a function of arc length: $t(s) = \frac{s}{\sqrt{a^2\omega^2+v_z^2}}$.

- **Substitution into Cartesian Coordinates:**

$$x(s) = a\cos\left(\frac{\omega s}{\sqrt{a^2\omega^2 + v_z^2}}\right)$$

$$y(s) = a\sin\left(\frac{\omega s}{\sqrt{a^2\omega^2 + v_z^2}}\right)$$

$$z(s) = \frac{v_z s}{\sqrt{a^2\omega^2 + v_z^2}}$$

(g) Center of Curvature C at $(a, 0, 0)$

> **Pedagogical Justification:** The center of curvature C of the osculating circle is displaced from particle P along the principal normal $\mathbf{e}_n$ by the radius of curvature ρ (Notes Sec. 2.4, Fig. 2.2):

$$\mathbf{r}_C = \mathbf{r}_P + \rho\mathbf{e}_n$$

- **Evaluation at $(a, 0, 0)$:**

At position $(a, 0, 0)$, $x(t) = a\cos(\omega t) = a \implies \omega t = 0 \implies t = 0$.

At $t = 0$:

- Position: $\mathbf{r}_P(0) = a\mathbf{i}_0$.
- Unit normal: $\mathbf{e}_n(0) = -\mathbf{e}_R(0) = -(\cos(0)\mathbf{i}_0 + \sin(0)\mathbf{j}_0) = -\mathbf{i}_0$.
- Radius of curvature: $\rho = a + \frac{v_z^2}{a\omega^2}$.

- **Algebraic Combination:**

$$\mathbf{r}_C = a\mathbf{i}_0 + \left(a + \frac{v_z^2}{a\omega^2}\right)(-\mathbf{i}_0) = a\mathbf{i}_0 - a\mathbf{i}_0 - \frac{v_z^2}{a\omega^2}\mathbf{i}_0 = -\frac{v_z^2}{a\omega^2}\mathbf{i}_0$$

In Cartesian coordinates:

$$C = \left(-\frac{v_z^2}{a\omega^2},\, 0,\, 0\right)$$

PHASE 4

Physical Interpretation & Dimensional Verification

- **Dimensional Check:**
- Velocity: $[a\omega] = [\mathrm{m}][\mathrm{s}^{-1}] = \mathrm{m/s}$, $[v_z] = \mathrm{m/s}$. $\implies$ $\sqrt{a^2\omega^2 + v_z^2}$ has units $\mathrm{m/s}$ (Consistent).
- Curvature center offset: $\left[\frac{v_z^2}{a\omega^2}\right] = \frac{[\mathrm{m/s}]^2}{[\mathrm{m}][\mathrm{s}^{-1}]^2} = \frac{\mathrm{m}^2/\mathrm{s}^2}{\mathrm{m/s}^2} = \mathrm{m}$ (Consistent).
- **Limiting Asymptotic Cases:**
- If $v_z \to 0$ (planar circular motion): $\rho = a$, and $C = (0, 0, 0)$, the origin of the circle as expected physically.
- If $\omega \to 0$ (pure vertical line ascent): $\rho \to \infty$, curvature vanishes, and the trajectory becomes a straight vertical line.

Source: Problems.pdf (Taken from MIT OCW)

PROBLEM 02

Kinematics, Spiral

Original statement · Problems.pdf, p. 6

Polar Kinematics

A planar spiral is given by the equation $r = \alpha e^{\beta\theta}$, where α and β are constants, r is the radial coordinate, and θ is the angular coordinate. If a particle moves along the spiral according to the equation $\theta = \delta t^n$ (where δ is a constant), determine:

(a) The radial and transverse components of the velocity.

(b) The radial and transverse components of the acceleration.

(c) The tangential and normal components of the velocity.

(d) The tangential and normal components of the acceleration.

(e) The radius of curvature of the spiral.

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** A point particle P moves along a logarithmic/exponential planar spiral $r(\theta) = \alpha e^{\beta\theta}$ with a prescribed power-law angular motion $\theta(t) = \delta t^n$.
- **Hypotheses:** $\alpha, \beta, \delta > 0$ are positive constants, $n > 0$ is an integer or real exponent, $t \geq 0$. Motion is strictly 2D planar in the Oxy plane ($z = 0$).
- **Degrees of Freedom:** $\mathrm{CDOF} = 1$ parameterized by time t.
- **Parameters & Units:** $[r] = \mathrm{m} \implies [\alpha] = \mathrm{m}$, β is dimensionless ($[-]$), $[\theta] = \mathrm{rad} \implies [\delta] = \mathrm{rad} \cdot \mathrm{s}^{-n}$.

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Inertial Basis:** $\mathcal{B}_0 = \{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$.
- **Polar Basis:** $\mathcal{B}_{\mathrm{polar}} = \{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}_0\}$.
- **Rotation Matrix $[{}_0R_\theta]$:**

$$[{}_0R_\theta] = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} \mathbf{e}_r \\ \mathbf{e}_\theta \\ \mathbf{k}_0 \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \mathbf{i}_0 \\ \mathbf{j}_0 \\ \mathbf{k}_0 \end{pmatrix}$$

- **Time Derivatives of Polar Unit Vectors (Notes Sec. 2.6.2, Eqs. 2.25–2.26):**

$$\dot{\mathbf{e}}_r = \dot\theta \mathbf{e}_\theta, \quad \dot{\mathbf{e}}_\theta = -\dot\theta \mathbf{e}_r$$

- **Orthonormality of $[{}_0R_\theta]$:** $\det[{}_0R_\theta] = \cos^2\theta + \sin^2\theta = 1$ and $[{}_0R_\theta][{}_0R_\theta]^T = I$ (checked symbolically with sympy).

- **Intrinsic basis from the polar basis.** Since $\mathbf{v} = \dot r\,\mathbf{e}_r + r\dot\theta\,\mathbf{e}_\theta = r\dot\theta\,(\beta\,\mathbf{e}_r + \mathbf{e}_\theta)$ (Phase 3, part (a)), the unit tangent is $\mathbf{e}_t = (\beta\,\mathbf{e}_r + \mathbf{e}_\theta)/\sqrt{1+\beta^2}$ and the in-plane unit normal is $\mathbf{e}_n = \mathbf{k}_0 \times \mathbf{e}_t = (-\mathbf{e}_r + \beta\,\mathbf{e}_\theta)/\sqrt{1+\beta^2}$. Their components in $\{\mathbf{e}_r, \mathbf{e}_\theta\}$ are the columns of

$$[{}_{\text{polar}}R_{\text{tn}}] = \frac{1}{\sqrt{1+\beta^2}}\begin{pmatrix}\beta & -1\\ 1 & \beta\end{pmatrix}, \qquad \det = \frac{\beta^2+1}{1+\beta^2} = 1, \qquad RR^T = I$$

- This is a planar rotation by the constant angle $\psi = \arctan(1/\beta)$ (the equiangular property of the spiral). Sympy confirms that $\mathbf{a}\cdot\mathbf{e}_t$ and $\mathbf{a}\cdot\mathbf{e}_n$ reproduce the a_t and a_n of Phase 3, parts (d), with $a_n > 0$, so $\mathbf{e}_n$ points to the concave side.

PHASE 3

Step-by-Step Mathematical Deduction

(a) Radial and Transverse Velocity Components

Pedagogical Justification: In polar coordinates, the velocity vector is given by $\mathbf{v} = \dot r\mathbf{e}_r + r\dot\theta\mathbf{e}_\theta$. The radial component $v_r = \dot r$ measures the expansion rate, while $v_\theta = r\dot\theta$ measures transverse rotation.

- **Explicit Chain Rule for $\dot\theta(t)$ and $\dot r(t)$:**

$$\theta(t) = \delta t^n \implies \dot\theta(t) = \frac{d}{dt}(\delta t^n) = n\delta t^{n-1}$$

Using the chain rule $\dot r = \frac{dr}{d\theta}\dot\theta$:

$$\frac{dr}{d\theta} = \frac{d}{d\theta}(\alpha e^{\beta\theta}) = \alpha\beta e^{\beta\theta} = \beta r$$

$$\dot r = (\beta r)(n\delta t^{n-1}) = \beta n\delta t^{n-1} r = \alpha\beta n\delta t^{n-1} e^{\beta\delta t^n}$$

- **Components:**

$$v_r = \dot{r} = \beta n \delta t^{n-1} r$$

$$v_\theta = r\dot{\theta} = n \delta t^{n-1} r$$

(b) Radial and Transverse Acceleration Components

> **Pedagogical Justification:** The acceleration in polar coordinates (Notes Eq. 2.30) accounts for the curvature of the polar grid:

$$\mathbf{a} = a_r \mathbf{e}_r + a_\theta \mathbf{e}_\theta = (\ddot{r} - r\dot{\theta}^2)\mathbf{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\mathbf{e}_\theta$$

- **Second Time Derivatives (Explicit Product Rule):**

$$\ddot{\theta}(t) = \frac{d}{dt}(n\delta t^{n-1}) = n(n-1)\delta t^{n-2}$$

For $\ddot{r}(t) = \frac{d}{dt}(\beta r \dot{\theta})$:

$$\ddot{r} = \beta(\dot{r}\dot{\theta} + r\ddot{\theta}) = \beta\left((\beta r\dot{\theta})\dot{\theta} + r\ddot{\theta}\right) = \beta^2 r\dot{\theta}^2 + \beta r\ddot{\theta}$$

- **Radial Acceleration a_r:**

$$a_r = \ddot{r} - r\dot{\theta}^2 = (\beta^2 r\dot{\theta}^2 + \beta r\ddot{\theta}) - r\dot{\theta}^2 = (\beta^2 - 1)r\dot{\theta}^2 + \beta r\ddot{\theta}$$

Substituting $\dot{\theta} = n\delta t^{n-1}$ and $\ddot{\theta} = n(n-1)\delta t^{n-2}$:

$$a_r = (\beta^2 - 1) r (n\delta t^{n-1})^2 + \beta r n(n-1)\delta t^{n-2} = n\delta t^{n-2} r \left[(\beta^2 - 1) n \delta t^n + \beta(n-1) \right]$$

- **Transverse Acceleration a_θ:**

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = r\ddot{\theta} + 2(\beta r \dot{\theta})\dot{\theta} = r\ddot{\theta} + 2\beta r \dot{\theta}^2$$

Substituting derivatives:

$$a_\theta = rn(n-1)\delta t^{n-2} + 2\beta r (n\delta t^{n-1})^2 = n\delta t^{n-2} r \left[(n-1) + 2\beta n \delta t^n \right]$$

(c) Tangential and Normal Velocity Components

> **Pedagogical Justification:** In the intrinsic Frenet basis, the velocity vector is strictly tangential: $\mathbf{v} = v\mathbf{e}_t = v_t \mathbf{e}_t + 0\mathbf{e}_n$. Therefore, the normal velocity component is identically zero.

- **Calculation:**

$$v_t = v = \sqrt{v_r^2 + v_\theta^2} = \sqrt{(\beta r \dot{\theta})^2 + (r\dot{\theta})^2} = r\dot{\theta}\sqrt{\beta^2 + 1} = n\delta t^{n-1} r \sqrt{1 + \beta^2}$$

$$v_n = 0$$

(d) Tangential and Normal Acceleration Components

> **Pedagogical Justification:** Tangential acceleration $a_t = \dot{v}$ measures the rate of change of speed, while normal acceleration $a_n = \frac{v^2}{\rho}$ measures directional turning.

- **Tangential Acceleration a_t (Explicit Differentiation):**

$$a_t = \frac{d}{dt}(v) = \sqrt{1+\beta^2}\frac{d}{dt}(r\dot{\theta}) = \sqrt{1+\beta^2}(\dot{r}\dot{\theta} + r\ddot{\theta})$$

Substituting $\dot{r} = \beta r\dot{\theta}$:

$$a_t = \sqrt{1+\beta^2}(\beta r\dot{\theta}^2 + r\ddot{\theta}) = \sqrt{1+\beta^2}n\delta t^{n-2}r\left[\beta n\delta t^n + (n-1)\right]$$

- **Normal Acceleration a_n:**

Using the Pythagorean decomposition of the total acceleration magnitude: $\|\mathbf{a}\|^2 = a_t^2 + a_n^2 = a_r^2 + a_\theta^2$:

$$a_n = \sqrt{a_r^2 + a_\theta^2 - a_t^2}$$

Alternatively, using the cross product $\mathbf{a} \times \mathbf{v} = (a_t\mathbf{e}_t + a_n\mathbf{e}_n) \times (v\mathbf{e}_t) = -va_n\mathbf{e}_b$:

$$a_n = \frac{\|\mathbf{v} \times \mathbf{a}\|}{v} = \frac{|v_r a_\theta - v_\theta a_r|}{v}$$

Computing the numerator:

$$v_r a_\theta - v_\theta a_r = (\beta r\dot{\theta})(r\ddot{\theta} + 2\beta r\dot{\theta}^2) - (r\dot{\theta})((\beta^2-1)r\dot{\theta}^2 + \beta r\ddot{\theta})$$

$$= \beta r^2\dot{\theta}\ddot{\theta} + 2\beta^2 r^2\dot{\theta}^3 - (\beta^2-1)r^2\dot{\theta}^3 - \beta r^2\dot{\theta}\ddot{\theta} = (2\beta^2 - \beta^2 + 1)r^2\dot{\theta}^3 = (\beta^2+1)r^2\dot{\theta}^3$$

Dividing by $v = r\dot{\theta}\sqrt{\beta^2+1}$:

$$a_n = \frac{(\beta^2+1)r^2\dot{\theta}^3}{r\dot{\theta}\sqrt{\beta^2+1}} = r\dot{\theta}^2\sqrt{\beta^2+1} = n^2\delta^2 t^{2n-2} r\sqrt{1+\beta^2}$$

(e) Radius of Curvature of the Spiral

Pedagogical Justification: The radius of curvature is obtained directly from $a_n = \frac{v^2}{\rho} \implies \rho = \frac{v^2}{a_n}$.

- **Calculation:**

$$\rho = \frac{v^2}{a_n} = \frac{(r\dot{\theta}\sqrt{1+\beta^2})^2}{r\dot{\theta}^2\sqrt{1+\beta^2}} = \frac{r^2\dot{\theta}^2(1+\beta^2)}{r\dot{\theta}^2\sqrt{1+\beta^2}} = r\sqrt{1+\beta^2}$$

PHASE 4

Physical Interpretation & Dimensional Verification

- **Geometric Invariance:** In a logarithmic spiral $r = \alpha e^{\beta\theta}$, the angle ψ between the radial line and the tangent vector satisfies $\tan\psi = \frac{r}{dr/d\theta} = \frac{r}{\beta r} = \frac{1}{\beta} = \text{constant}$. This is the celebrated equiangular spiral property!
- **Radius of Curvature:** $\rho = r\sqrt{1+\beta^2}$ is strictly proportional to current radial distance r.
- **Units Check:** $[\rho] = [r][\sqrt{1+\beta^2}] = \text{m} \cdot 1 = \text{m}$ (Consistent).
- **Limiting Case $\beta \to 0$:** The spiral degenerates into a pure circle $r = \alpha$, where $\rho = r\sqrt{1+0} = r$, recovering the circle radius of curvature.

Source: Problems.pdf (Page 6)

PROBLEM 03

Kinematics, Astroid

Original statement · Problems.pdf, p. 6

Intrinsic Kinematics

A point particle describes the astroid:

$$x = a\cos^3\phi, \qquad y = a\sin^3\phi$$

with a motion whose tangential acceleration is constant, $a_t = k$. If the point is initially at rest at $\phi = 0$, compute the evolution of the position as a function of time while the particle is in one of the smooth segments of the astroid. *(Hint:* $ds = \sqrt{dx^2 + dy^2}$*).*

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** A point particle P moves along a hypocycloid with four cusps (an astroid), parameterized by angle ϕ.
- **Hypotheses:** The motion starts from rest at $\phi(0) = 0$ with constant tangential acceleration $a_t = k = \text{const} > 0$. The particle remains within the first quadrant segment $0 \le \phi \le \pi/2$.
- **Degrees of Freedom:** $\text{CDOF} = 1$.
- **Parameters & Units:** Astroid scale a $[\text{m}]$, tangential acceleration k $[\text{m/s}^2]$, time t $[\text{s}]$.

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Inertial Cartesian Frame:** $\mathcal{S}_0 = \{O; \mathbf{i}_0, \mathbf{j}_0\}$.
- **Parametric Equations of the Astroid:**

$$x(\phi) = a\cos^3\phi, \quad y(\phi) = a\sin^3\phi$$

- **Metric Arc Length Differential** ds**:**

Differentiating with respect to parameter ϕ:

$$dx = \frac{dx}{d\phi}d\phi = 3a\cos^2\phi(-\sin\phi)d\phi = -3a\cos^2\phi\sin\phi\,d\phi$$

$$dy = \frac{dy}{d\phi}d\phi = 3a\sin^2\phi(\cos\phi)d\phi = 3a\sin^2\phi\cos\phi\,d\phi$$

The squared differential arc length is:

$$ds^2 = dx^2 + dy^2 = 9a^2\cos^4\phi\sin^2\phi\,d\phi^2 + 9a^2\sin^4\phi\cos^2\phi\,d\phi^2$$

Factoring out $9a^2\cos^2\phi\sin^2\phi$:

$$ds^2 = 9a^2\cos^2\phi\sin^2\phi(\cos^2\phi + \sin^2\phi)d\phi^2 = 9a^2\cos^2\phi\sin^2\phi\,d\phi^2$$

Taking the positive square root for $0 \le \phi \le \pi/2$:

$$ds = 3a\sin\phi\cos\phi\,d\phi = \frac{3a}{2}\sin(2\phi)\,d\phi$$

PHASE 3

Step-by-Step Mathematical Deduction

Step 1: Intrinsic Kinematics from Constant Tangential Acceleration $a_t = k$

Pedagogical Justification: Tangential acceleration is the time derivative of speed: $a_t = \frac{dv}{dt} = \ddot{s}$. Integrating once with initial condition $v(0) = 0$:

$$v(t) = \dot{s}(t) = \int_0^t a_t\,dt' = \int_0^t k\,dt' = kt$$

- **Second Integration for Arc Length $s(t)$:**

Integrating $v(t) = \dot{s}(t)$ with initial condition $s(0) = 0$ (at $\phi = 0$):

$$s(t) = \int_0^t kt'\,dt' = \left[\frac{1}{2}kt'^2\right]_0^t = \frac{1}{2}kt^2$$

Step 2: Arc Length as a Function of Parameter ϕ

Pedagogical Justification: We integrate the geometric differential $ds = 3a\sin u\cos u\,du$ from the origin $\phi = 0$ to current ϕ:

- **Explicit Substitution Integration:**

Let $w = \sin u \implies dw = \cos u\,du$. When $u = 0, w = 0$; when $u = \phi, w = \sin\phi$:

$$s(\phi) = \int_0^{\phi} 3a\sin u\cos u\,du = 3a\int_0^{\sin\phi} w\,dw = 3a\left[\frac{w^2}{2}\right]_0^{\sin\phi} = \frac{3a}{2}\sin^2\phi$$

Step 3: Equating Kinematic and Geometric Arc Length

> **Pedagogical Justification:** The distance traveled along the curve $s(t)$ must equal the geometric arc length $s(\phi)$:

$$\frac{3a}{2}\sin^2\phi = \frac{1}{2}kt^2$$

Multiplying both sides by $\frac{2}{3a}$:

$$\sin^2\phi = \frac{kt^2}{3a}$$

Taking the positive square root (since $0 \le \phi \le \pi/2$):

$$\sin\phi = \sqrt{\frac{k}{3a}}\,t$$

Applying the inverse sine function:

$$\phi(t) = \arcsin\left(\sqrt{\frac{k}{3a}}\,t\right)$$

(Compare with the official solution key (Problems.pdf, Solution keys section): $\phi = \arcsin\sqrt{\frac{k}{3a}t^2} = \arcsin(\sqrt{\frac{k}{3a}}t)$.)

Step 4: Evolution of the Cartesian Position as a Function of Time

Pedagogical Justification: Substituting $\phi(t)$ into the parametric definitions of the astroid:

Using trigonometric identities: $\sin\phi = \sqrt{\frac{k}{3a}}t$ and $\cos\phi = \sqrt{1-\sin^2\phi} = \sqrt{1-\frac{kt^2}{3a}}$:

$$x(t) = a\cos^3\phi = a\left(1-\frac{kt^2}{3a}\right)^{3/2}$$

$$y(t) = a\sin^3\phi = a\left(\frac{kt^2}{3a}\right)^{3/2} = a\left(\frac{k}{3a}\right)^{3/2}t^3$$

PHASE 4

Physical Interpretation & Dimensional Verification

- **Dimensional Check:**
- In $\sin\phi = \sqrt{\frac{k}{3a}}t$:

$$\left[\sqrt{\frac{k}{3a}}\,t\right] = \sqrt{\frac{\mathrm{m/s^2}}{\mathrm{m}}}\,\mathrm{s} = \sqrt{\mathrm{s^{-2}}}\,\mathrm{s} = \mathrm{s^{-1}}\cdot\mathrm{s} = 1 \quad \text{(Dimensionless argument for arcsin! ?)}$$

- **Physical Limits:**
- At $t=0$: $\phi = 0 \implies x(0) = a, y(0) = 0$ (starts at the right cusp).
- The particle reaches the top cusp $\phi = \pi/2$ when $\sin\phi = 1$:

$$\sqrt{\frac{k}{3a}}t_{\text{cusp}} = 1 \implies t_{\text{cusp}} = \sqrt{\frac{3a}{k}}$$

At $t_{\text{cusp}}, x(t_{\text{cusp}}) = 0, y(t_{\text{cusp}}) = a.$

Source: Problems.pdf (Page 6)

PROBLEM 04

Plane at Constant Altitude

Original statement · Problems.pdf, p. 7

Coordinate Transformations

An aircraft moves at constant altitude y_0 with velocity v_0 in the x direction. At time $t = 0$ it is located at $x = 0$, and it begins to accelerate in the x direction at a constant rate a_x. Describe the position, velocity, and acceleration in polar coordinates r, θ, and answer the following questions:

(a) Show that: $\dot{r} = \frac{v(t)x(t)}{r(t)}$

(b) Show that: $\dot{\theta} = -\frac{v(t)y_0}{r^2(t)}$

(c) Show that: $\ddot{r} = \frac{v^2(t)y_0^2 + a_x x(t) r^2(t)}{r^3(t)}$

(d) Show that: $\ddot{\theta} = y_0 \frac{2v^2(t)x(t) - a_x r^2(t)}{r^4(t)}$

From these expressions form the acceleration vector $\mathbf{a} = a_r \mathbf{e}_r + a_\theta \mathbf{e}_\theta$. Repeat the problem using Cartesian coordinates.

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** An aircraft flies at constant altitude y_0 with velocity in the x direction. At $t = 0$, $x(0) = 0$ and velocity is v_0. It accelerates in the x direction at constant rate a_x.
- **Tracking System:** A ground radar station located at origin $O(0, 0)$ tracks the aircraft using range $r(t)$ and elevation/bearing angle $\theta(t)$.
- **Hypotheses:** $y_0 > 0, v_0 \geq 0, a_x = \text{const}$. Altitude is fixed: $y(t) \equiv y_0 \implies \dot{y} = 0, \ddot{y} = 0$.

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Inertial Cartesian Frame:** Coordinates (x, y) with position $\mathbf{r} = x\mathbf{i} + y_0\mathbf{j}$.
- **Radar Polar Frame:** Position $\mathbf{r} = r\mathbf{e}_r$, where:

$$r(t) = \sqrt{x(t)^2 + y_0^2}, \quad \tan\theta(t) = \frac{y_0}{x(t)} \implies \theta(t) = \arctan\left(\frac{y_0}{x(t)}\right)$$

- **Kinematics in x:**

$$\dot{x}(t) = v(t) = v_0 + a_x t, \quad \ddot{x}(t) = a_x$$

$$x(t) = v_0 t + \frac{1}{2} a_x t^2$$

- **Change of basis (radar polar basis).** With θ the angle of the line of sight measured from $+\mathbf{i}$, the polar basis $\mathcal{B}_1 = \{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}\}$ is the inertial basis $\mathcal{B}_0 = \{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ rotated by $\theta(t)$ about $\mathbf{k}$. The columns are the components of $\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}$ in $\mathcal{B}_0$; using $\cos\theta = x/r$ and

$\sin\theta = y_0/r$:

$$[{}_0R_1] = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} x/r & -y_0/r & 0 \\ y_0/r & x/r & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\det[{}_0R_1] = \frac{x^2 + y_0^2}{r^2} = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

- Both properties were verified symbolically with sympy. The inverse relation $\mathbf{i} = \cos\theta\,\mathbf{e}_r - \sin\theta\,\mathbf{e}_\theta$ (first row of $[{}_0R_1]^T$ applied to the basis) is the one used in the Cartesian cross-check of Phase 3.

PHASE 3

Step-by-Step Mathematical Deduction

(a) Proof that $\dot{r} = \frac{v(t)x(t)}{r(t)}$

Pedagogical Justification: Starting from the geometric relation $r(t)^2 = x(t)^2 + y_0^2$, we differentiate implicitly with respect to time t.

- **Implicit Differentiation:**

$$\frac{d}{dt}[r^2] = \frac{d}{dt}[x^2 + y_0^2]$$

Applying the chain rule $\frac{d}{dt}(r^2) = 2r\dot{r}$ and $\frac{d}{dt}(x^2) = 2x\dot{x}$:

$$2r\dot{r} = 2x\dot{x} + 0 = 2xv(t)$$

Dividing by $2r$:

$$\dot{r} = \frac{v(t)x(t)}{r(t)} \quad \blacksquare$$

(b) Proof that $\dot{\theta} = -\frac{v(t)y_0}{r(t)^2}$

> **Pedagogical Justification:** Starting from $\tan\theta = \frac{y_0}{x}$, we differentiate both sides with respect to time using the quotient rule.

- **Differentiation:**

$$\frac{d}{dt}(\tan\theta) = \frac{d}{dt}\left(\frac{y_0}{x}\right)$$

Recalling $\frac{d}{dt}(\tan\theta) = (1+\tan^2\theta)\dot{\theta} = \sec^2\theta\,\dot{\theta}$, and $\frac{d}{dt}(x^{-1}) = -\frac{\dot{x}}{x^2} = -\frac{v}{x^2}$:

$$\sec^2\theta\,\dot{\theta} = -\frac{y_0 v(t)}{x(t)^2}$$

From geometry: $\cos\theta = \frac{x}{r} \implies \sec\theta = \frac{r}{x} \implies \sec^2\theta = \frac{r^2}{x^2}$:

$$\frac{r^2}{x^2}\dot{\theta} = -\frac{y_0 v(t)}{x^2}$$

Multiplying both sides by x^2 and dividing by r^2:

$$\dot{\theta} = -\frac{v(t)y_0}{r(t)^2} \quad \blacksquare$$

(c) Proof that $\ddot{r} = \frac{v^2(t)y_0^2 + a_x x(t) r^2(t)}{r(t)^3}$

> **Pedagogical Justification:** We differentiate the expression for $\dot{r} = \frac{vx}{r}$ using the quotient rule $\left(\frac{u}{w}\right)' = \frac{u'w - uw'}{w^2}$.

- **Differentiation:**

Here $u = vx \implies \dot{u} = \dot{v}x + v\dot{x} = a_x x + v^2$.

The denominator is $w = r \implies \dot{w} = \dot{r} = \frac{vx}{r}$.

$$\ddot{r} = \frac{\dot{u}r - u\dot{r}}{r^2} = \frac{(a_x x + v^2)r - (vx)\left(\frac{vx}{r}\right)}{r^2}$$

Multiplying numerator and denominator by r:

$$\ddot{r} = \frac{(a_x x + v^2)r^2 - v^2 x^2}{r^3}$$

Expanding the numerator:

$$(a_x x + v^2)r^2 - v^2 x^2 = a_x x r^2 + v^2 r^2 - v^2 x^2 = a_x x r^2 + v^2(r^2 - x^2)$$

Recalling that $r^2 - x^2 = y_0^2$:

$$\ddot{r} = \frac{a_x x(t) r^2(t) + v^2(t) y_0^2}{r(t)^3} = \frac{v^2(t)y_0^2 + a_x x(t) r^2(t)}{r(t)^3} \quad \blacksquare$$

(d) Proof that $\ddot{\theta} = y_0 \frac{2v^2(t)x(t) - a_x r^2(t)}{r(t)^4}$

> **Pedagogical Justification:** We differentiate $\dot{\theta} = -y_0 v r^{-2}$ with respect to time using the product rule.

- **Differentiation:**

$$\ddot{\theta} = \frac{d}{dt}\left(-y_0 \frac{v}{r^2}\right) = -y_0 \frac{\dot{v}r^2 - v\frac{d}{dt}(r^2)}{r^4}$$

Substituting $\dot{v} = a_x$ and $\frac{d}{dt}(r^2) = 2r\dot{r} = 2r\left(\frac{vx}{r}\right) = 2vx$:

$$\ddot{\theta} = -y_0 \frac{a_x r^2 - v(2vx)}{r^4} = -y_0 \frac{a_x r^2 - 2v^2 x}{r^4} = y_0 \frac{2v^2(t)x(t) - a_x r^2(t)}{r(t)^4} \quad \blacksquare$$

Polar and Cartesian Acceleration Vectors

- **Polar Acceleration Vector:**

$$a_r = \ddot{r} - r\dot{\theta}^2 = \frac{v^2 y_0^2 + a_x x r^2}{r^3} - r\left(-\frac{v y_0}{r^2}\right)^2 = \frac{v^2 y_0^2 + a_x x r^2}{r^3} - \frac{v^2 y_0^2}{r^3} = \frac{a_x x r^2}{r^3} = a_x \frac{x}{r}$$

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = r\left(y_0 \frac{2v^2 x - a_x r^2}{r^4}\right) + 2\left(\frac{vx}{r}\right)\left(-\frac{v y_0}{r^2}\right) = \frac{2v^2 x y_0 - a_x y_0 r^2}{r^3} - \frac{2v^2 x y_0}{r^3} = -a_x \frac{y_0}{r}$$

$$\mathbf{a} = a_x\left(\frac{x}{r}\mathbf{e}_r - \frac{y_0}{r}\mathbf{e}_\theta\right)$$

- **Cartesian Verification:**

In Cartesian coordinates:

$$\mathbf{a} = \ddot{x}\mathbf{i}_0 + \ddot{y}\mathbf{j}_0 = a_x\mathbf{i}_0 + 0\mathbf{j}_0 = a_x\mathbf{i}_0$$

Converting Cartesian $\mathbf{i}_0$ into polar: $\mathbf{i}_0 = \cos\theta\mathbf{e}_r - \sin\theta\mathbf{e}_\theta = \frac{x}{r}\mathbf{e}_r - \frac{y_0}{r}\mathbf{e}_\theta$.

Substituting gives identically $\mathbf{a} = a_x\left(\frac{x}{r}\mathbf{e}_r - \frac{y_0}{r}\mathbf{e}_\theta\right)$. The match is mathematically exact!

PHASE 4

Physical Interpretation, Limiting Cases and Dimensional Check

- **Dimensional check:**
- $[\dot{r}] = [v\,x/r] = \mathrm{m/s}$; $\quad [\dot{\theta}] = [v\,y_0/r^2] = \dfrac{(\mathrm{m/s})\,\mathrm{m}}{\mathrm{m}^2} = \mathrm{s}^{-1}$ (rad/s).
- $[\ddot{r}] = \dfrac{(\mathrm{m}^2/\mathrm{s}^2)\,\mathrm{m}^2}{\mathrm{m}^3} = \mathrm{m/s}^2$ and $\dfrac{(\mathrm{m/s}^2)\,\mathrm{m}\,\mathrm{m}^2}{\mathrm{m}^3} = \mathrm{m/s}^2$; $\quad [\ddot{\theta}] = \dfrac{\mathrm{m}\,(\mathrm{m}^2/\mathrm{s}^2)\,\mathrm{m}}{\mathrm{m}^4} = \mathrm{s}^{-2}$ (Consistent).
- **Limit $x \to \infty$ (aircraft far away):** $r \to x$, so $\dot{r} \to v, \dot{\theta} \to 0, a_r \to a_x, a_\theta \to 0$: the line of sight aligns with the flight direction and the motion is purely radial.
- **Limit $x = 0$ (aircraft overhead, $t = 0$):** $\dot{r} = 0, \dot{\theta} = -v_0/y_0, a_r = 0, a_\theta = -a_x$. Then $\theta = \pi/2$ and $\mathbf{a} = a_x\mathbf{i} = -a_x\mathbf{e}_\theta$, as required by the change of basis.
- **Invariance of the modulus:** $a_r^2 + a_\theta^2 = a_x^2\,\dfrac{x^2 + y_0^2}{r^2} = a_x^2$, equal to $\|a_x\mathbf{i}\|^2$ because $[{}_0R_1]$ is orthogonal.
- **Symbolic check (sympy):** differentiating $r = \sqrt{x^2 + y_0^2}$ and $\theta = \arctan(y_0/x)$ with $x = v_0 t + \frac{1}{2}a_x t^2$ reproduces the four identities (a)-(d) with zero residual, and gives $a_r = a_x x/r, a_\theta = -a_x y_0/r$ with zero residual; the three limits above were obtained with sympy as well.

Source: Problems.pdf (Page 7)

PROBLEM 05

Plane and Glider

Original statement · Problems.pdf, p. 7

Aerospace Towed Flight

An airplane A is flying horizontally with a constant speed v_A, and is towing a glider B. The glider is gaining altitude. The tow cable has a length r, and the angle θ formed between the tow cable and the horizontal is increasing at a constant rate of $\dot{\theta}$. We want to determine the magnitude of the velocity, v_B, and the acceleration, a_B, of the glider as a function of θ and its derivatives.

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** Tow plane A flies horizontally at constant speed v_A. Glider B is towed behind on a cable of constant length r. The tow angle $\theta(t)$ above the horizontal increases at rate $\dot{\theta}(t)$.
- **Hypotheses:** Cable length $r = \text{const}$, plane speed $v_A = \text{const}$ (so $\dot{v}_A = 0, \mathbf{a}_A = \mathbf{0}$).
- **Parameters & Units:** Plane speed v_A $[\mathrm{m/s}]$, cable length r $[\mathrm{m}]$, angle θ $[\mathrm{rad}]$, angular rate $\dot{\theta}$ $[\mathrm{rad/s}]$, angular acceleration $\ddot{\theta}$ $[\mathrm{rad/s^2}]$.

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Inertial Reference Frame:** $\mathcal{S}_0 = \{O; \mathbf{i}_0, \mathbf{j}_0\}$, with $\mathbf{i}_0$ pointing horizontally in plane A's direction of flight, and $\mathbf{j}_0$ pointing vertically upward.
- **Position of Plane A:**

$$\mathbf{r}_A(t) = (x_{A0} + v_A t)\mathbf{i}_0$$

- **Relative Position of Glider B with Respect to A:**

Since the glider is behind A and gaining altitude while θ increases, the vector from A to B points at angle θ above the negative horizontal:

$$\mathbf{r}_{B/A} = -r\cos\theta\,\mathbf{i}_0 + r\sin\theta\,\mathbf{j}_0$$

The total position of glider B in $\mathcal{S}_0$ is:

$$\mathbf{r}_B = \mathbf{r}_A + \mathbf{r}_{B/A} = (x_{A0} + v_A t - r\cos\theta)\mathbf{i}_0 + (r\sin\theta)\mathbf{j}_0$$

Erratum: The first version said that the vector from A to B is at angle θ below the negative horizontal, which contradicts $\mathbf{r}_{B/A} = -r\cos\theta\,\mathbf{i}_0 + r\sin\theta\,\mathbf{j}_0$ (positive vertical component). The glider is behind and above A, at angle θ above the negative horizontal; this is the only reading compatible with the glider gaining altitude as θ increases, and the formulas for $\mathbf{v}_{B/0}$ and $\mathbf{a}_{B/0}$ are unchanged. Numeric check: the altitude $r\sin\theta$ has derivative $r\dot{\theta}\cos\theta > 0$ for $0 < \theta < \pi/2$ and $\dot{\theta} > 0$.

PHASE 3

Step-by-Step Mathematical Deduction

Velocity of Glider B

Pedagogical Justification: The absolute velocity is obtained by differentiating position $\mathbf{r}_B$ in the inertial frame $\mathcal{S}_0$:

$$\mathbf{v}_{B/0} = \frac{d\mathbf{r}_B}{dt} = \frac{d}{dt}(x_{A0} + v_A t - r\cos\theta)\mathbf{i}_0 + \frac{d}{dt}(r\sin\theta)\mathbf{j}_0$$

- **Explicit Chain Rule:**

$$\frac{d}{dt}(-r\cos\theta) = -r(-\sin\theta)\dot{\theta} = r\dot{\theta}\sin\theta$$

$$\frac{d}{dt}(r\sin\theta) = r\cos\theta\,\dot{\theta}$$

Therefore:

$$\mathbf{v}_{B/0} = (v_A + r\dot{\theta}\sin\theta)\mathbf{i}_0 + (r\dot{\theta}\cos\theta)\mathbf{j}_0$$

- **Magnitude of Velocity $v_{B/0}$:**

$$v_{B/0} = \|\mathbf{v}_{B/0}\| = \sqrt{(v_A + r\dot{\theta}\sin\theta)^2 + (r\dot{\theta}\cos\theta)^2}$$

(Compare with the official solution key (Problems.pdf, Solution keys section).)

Acceleration of Glider B

> **Pedagogical Justification:** Acceleration is the time derivative of velocity $\mathbf{a}_{B/0} = \frac{d\mathbf{v}_{B/0}}{dt}$.

- **Explicit Product & Chain Rules for Each Component:**
- x-component differentiation:

$$\frac{d}{dt}(v_A + r\dot{\theta}\sin\theta) = 0 + r\frac{d}{dt}(\dot{\theta}\sin\theta) = r(\ddot{\theta}\sin\theta + \dot{\theta}(\cos\theta\dot{\theta})) = r\ddot{\theta}\sin\theta + r\dot{\theta}^2\cos\theta$$

- y-component differentiation:

$$\frac{d}{dt}(r\dot{\theta}\cos\theta) = r\frac{d}{dt}(\dot{\theta}\cos\theta) = r(\ddot{\theta}\cos\theta + \dot{\theta}(-\sin\theta\dot{\theta})) = r\ddot{\theta}\cos\theta - r\dot{\theta}^2\sin\theta$$

- **Vector Form:**

$$\mathbf{a}_{B/0} = (r\ddot{\theta}\sin\theta + r\dot{\theta}^2\cos\theta)\mathbf{i}_0 + (r\ddot{\theta}\cos\theta - r\dot{\theta}^2\sin\theta)\mathbf{j}_0$$

- **Magnitude of Acceleration $a_{B/0}$:**

$$a_{B/0}^2 = (r\ddot{\theta}\sin\theta + r\dot{\theta}^2\cos\theta)^2 + (r\ddot{\theta}\cos\theta - r\dot{\theta}^2\sin\theta)^2$$

Expanding both squared terms:

$$= r^2\ddot{\theta}^2\sin^2\theta + 2r^2\ddot{\theta}\dot{\theta}^2\sin\theta\cos\theta + r^2\dot{\theta}^4\cos^2\theta$$

$$+ r^2\ddot{\theta}^2\cos^2\theta - 2r^2\ddot{\theta}\dot{\theta}^2\sin\theta\cos\theta + r^2\dot{\theta}^4\sin^2\theta$$

The cross terms cancel out completely:

$$a_{B/0}^2 = r^2\ddot{\theta}^2(\sin^2\theta + \cos^2\theta) + r^2\dot{\theta}^4(\cos^2\theta + \sin^2\theta) = r^2\ddot{\theta}^2 + r^2\dot{\theta}^4$$

Taking the square root:

$$a_{B/0} = \sqrt{r^2\ddot{\theta}^2 + r^2\dot{\theta}^4}$$

(Compare with the official solution key (Problems.pdf, Solution keys section).)

PHASE 4

Physical Interpretation & Dimensional Verification

- **Plane Motion Decoupling:** Notice that the acceleration magnitude $a_{B/0}$ is completely independent of the tow plane's cruise velocity v_A because $\mathbf{a}_A = \mathbf{0}$ (uniform rectilinear translation). The acceleration felt by the glider is purely relative circular motion around A!
- **Limiting Case $\ddot{\theta} = 0$:** If the cable turns at a constant angular rate $\dot{\theta} = \Omega$, the acceleration reduces to pure centripetal acceleration: $a_{B/0} = \sqrt{r^2\Omega^4} = r\Omega^2$.
- **Dimensional check:** $[r\ddot{\theta}] = [r\dot{\theta}^2] = \mathrm{m}\cdot\mathrm{s}^{-2}$ and $[v_A + r\dot{\theta}\sin\theta] = \mathrm{m/s}$, so $a_{B/0} = \sqrt{r^2\ddot{\theta}^2 + r^2\dot{\theta}^4}$ is in $\mathrm{m/s^2}$ and $v_{B/0}$ in m/s.
- **Limit $\dot{\theta} = 0$:** $\mathbf{v}_{B/0} = v_A\,\mathbf{i}_0$ and $\mathbf{a}_{B/0} = \mathbf{0}$, the glider moves rigidly with the tow plane.
- **Symbolic check (sympy):** differentiating $\mathbf{r}_B$ twice for an arbitrary function $\theta(t)$ gives $\|\mathbf{v}_{B/0}\|^2 = (v_A + r\dot{\theta}\sin\theta)^2 + (r\dot{\theta}\cos\theta)^2$ and $\|\mathbf{a}_{B/0}\|^2 = r^2\ddot{\theta}^2 + r^2\dot{\theta}^4$ with zero residual.

Source: Problems.pdf (Page 7)

PROBLEM 06

Particle on Sinusoidal Curve

Original statement · Problems.pdf, p. 7

Frenet-Serret Frame

A point particle P travels along a curve given by:

$$y = \sin x$$

with a velocity along x given by $\dot{x} = 1$.

(a) Find the intrinsic, unitary vector basis associated to the curve.

(b) Express x as a function of an arc length parameter, s. You can leave your solution as a function of an integral.

(c) Find the tangent and normal acceleration components of P as a function of x.

(d) Write the radius of curvature of the trajectory as a function of x.

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** A particle travels along a sinusoidal path $y(x) = \sin x$ with constant horizontal speed $\dot{x} = 1$.
- **Hypotheses:** $\dot{x}(t) \equiv 1\ \text{m/s} \implies x(t) = t + x_0$ (taking $x_0 = 0 \implies x(t) = t$). $\ddot{x} = 0$.
- **Degrees of Freedom:** $\text{CDOF} = 1$.

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Parametric Curve:** $\mathbf{r}(x) = x\mathbf{i}_0 + \sin x\,\mathbf{j}_0$.
- **Derivative Vector:** $\frac{d\mathbf{r}}{dx} = \mathbf{i}_0 + \cos x\,\mathbf{j}_0$.
- **Differential Arc Length:**

$$ds = \left\| \frac{d\mathbf{r}}{dx} \right\| dx = \sqrt{1 + \cos^2 x}\, dx$$

- **Change of basis (intrinsic basis).** Let $\alpha(x) = \arctan(\cos x)$ be the slope angle of the tangent, so that $\mathbf{e}_t = (\cos\alpha, \sin\alpha) = (1, \cos x)/\sqrt{1+\cos^2 x}$. The in-plane normal used in this solution is $\mathbf{e}_n = \mathbf{k}_0 \times \mathbf{e}_t = (-\sin\alpha, \cos\alpha) = (-\cos x, 1)/\sqrt{1+\cos^2 x}$, the normal obtained by turning $\mathbf{e}_t$ by $+90^\circ$ (to its left). The columns are the components of $\mathbf{e}_t$, $\mathbf{e}_n$ in $\{\mathbf{i}_0, \mathbf{j}_0\}$:

$$[{}_0R_1] = \frac{1}{\sqrt{1+\cos^2 x}} \begin{pmatrix} 1 & -\cos x \\ \cos x & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \frac{1+\cos^2 x}{1+\cos^2 x} = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

- Because $\mathbf{e}_n$ is always the left normal, the sign of a_n tells on which side the centre of curvature lies: $a_n < 0$ (as for $0 < x < \pi$, where $\sin x > 0$) places it to the right of the direction of motion, below the arch; $a_n > 0$ for $\pi < x < 2\pi$ places it above the curve. The matrix properties were verified symbolically with sympy.

PHASE 3

Step-by-Step Mathematical Deduction

(a) Intrinsic Unitary Vector Basis $\{\mathbf{e}_t, \mathbf{e}_n\}$

- **Unit Tangent Vector $\mathbf{e}_t$:**

$$\mathbf{e}_t = \frac{d\mathbf{r}/dx}{\|d\mathbf{r}/dx\|} = \frac{\mathbf{i}_0 + \cos x\,\mathbf{j}_0}{\sqrt{1+\cos^2 x}}$$

- **Unit Principal Normal Vector $\mathbf{e}_n$:**

In a 2D plane, the normal vector $\mathbf{e}_n = \mathbf{k}_0 \times \mathbf{e}_t$ is perpendicular to $\mathbf{e}_t$ and oriented to its left (a $+90^\circ$ turn). It points toward the centre of curvature only where the signed curvature is positive.

Differentiating $\mathbf{e}_t$ with respect to arc length s:

$$\frac{d\mathbf{e}_t}{ds} = \frac{d\mathbf{e}_t}{dx}\frac{dx}{ds} = \frac{1}{\sqrt{1+\cos^2 x}}\frac{d}{dx}\left[(1+\cos^2 x)^{-1/2}(\mathbf{i}_0 + \cos x\,\mathbf{j}_0)\right]$$

Applying the quotient/product rule:

$$\frac{d}{dx}\left[\frac{1}{\sqrt{1+\cos^2 x}}\right] = -\frac{1}{2}(1+\cos^2 x)^{-3/2}(2\cos x(-\sin x)) = \frac{\sin x \cos x}{(1+\cos^2 x)^{3/2}}$$

$$\frac{d\mathbf{e}_t}{dx} = \frac{\sin x\cos x}{(1+\cos^2 x)^{3/2}}(\mathbf{i}_0 + \cos x\,\mathbf{j}_0) + \frac{1}{\sqrt{1+\cos^2 x}}(-\sin x\,\mathbf{j}_0)$$

Putting over the common denominator $(1+\cos^2 x)^{3/2}$:

$$= \frac{\sin x\cos x\,\mathbf{i}_0 + \sin x\cos^2 x\,\mathbf{j}_0 - \sin x(1+\cos^2 x)\,\mathbf{j}_0}{(1+\cos^2 x)^{3/2}} = \frac{\sin x\cos x\,\mathbf{i}_0 - \sin x\,\mathbf{j}_0}{(1+\cos^2 x)^{3/2}} = -\frac{\sin x}{(1+\cos^2 x)^{3/2}}(-\cos x\,\mathbf{i}_0 + \mathbf{j}_0)$$

Normalizing to unit length:

$$\mathbf{e}_n = \frac{-\cos x\,\mathbf{i}_0 + \mathbf{j}_0}{\sqrt{1+\cos^2 x}} \quad (\text{left normal } \mathbf{k}_0 \times \mathbf{e}_t)$$

Erratum: The first version called $\mathbf{e}_n$ the normal pointing to the concave side. It is the left normal $\mathbf{k}_0 \times \mathbf{e}_t$ and, since $d\mathbf{e}_t/ds = -\dfrac{\sin x}{(1+\cos^2 x)^{3/2}}\,\mathbf{e}_n$, it points away from the centre of curvature for $0 < x < \pi$ (where $a_n < 0$) and toward it for $\pi < x < 2\pi$ (where $a_n > 0$). Numeric check: sympy gives $a_n(\pi/2) = -1\ \mathrm{m/s^2}$, with the centre of curvature below the crest.

(b) Arc Length Parameter $s(x)$

Pedagogical Justification: The relation between x and arc length s is given by integrating the metric differential:

$$s(x) = \int_0^x \sqrt{1+\cos^2 u}\,du$$

This integral is an incomplete elliptic integral of the second kind, which cannot be expressed in elementary functions and is properly expressed as a quadrature.

(c) Tangent and Normal Acceleration Components

- **Velocity Vector:** Since $\dot{x} = 1$:

$$\mathbf{v} = \dot{x}\mathbf{i}_0 + \dot{y}\mathbf{j}_0 = 1\mathbf{i}_0 + \cos x\,\mathbf{j}_0$$

The speed is:

$$v(x) = \|\mathbf{v}\| = \sqrt{1 + \cos^2 x}$$

- **Acceleration Vector:** Differentiating velocity with $\dot{x} = 1$:

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \frac{d}{dt}(1\mathbf{i}_0 + \cos x\, \mathbf{j}_0) = -\sin x\, \dot{x}\, \mathbf{j}_0 = -\sin x\, \mathbf{j}_0$$

- **Tangential Acceleration Component a_t:**

$$a_t = \mathbf{a} \cdot \mathbf{e}_t = (-\sin x\, \mathbf{j}_0) \cdot \left(\frac{\mathbf{i}_0 + \cos x\, \mathbf{j}_0}{\sqrt{1 + \cos^2 x}} \right) = -\frac{\sin x \cos x}{\sqrt{1 + \cos^2 x}}$$

- **Normal Acceleration Component a_n:**

$$a_n = \mathbf{a} \cdot \mathbf{e}_n = (-\sin x\, \mathbf{j}_0) \cdot \left(\frac{-\cos x\, \mathbf{i}_0 + \mathbf{j}_0}{\sqrt{1 + \cos^2 x}} \right) = \frac{-\sin x}{\sqrt{1 + \cos^2 x}}$$

Its magnitude is:

$$|a_n| = \frac{|\sin x|}{\sqrt{1 + \cos^2 x}}$$

(d) Radius of Curvature $\rho(x)$

Pedagogical Justification: Using $a_n = \frac{v^2}{\rho}$:

$$\rho(x) = \frac{v^2}{|a_n|} = \frac{1 + \cos^2 x}{\frac{|\sin x|}{\sqrt{1+\cos^2 x}}} = \frac{(1 + \cos^2 x)^{3/2}}{|\sin x|}$$

(Notice this matches the standard Cartesian curvature formula $\rho = \frac{(1+y'^2)^{3/2}}{|y''|}$ with $y' = \cos x, y'' = -\sin x$!)

PHASE 4

Physical Interpretation, Limiting Cases and Dimensional Check

- **Dimensional check:** the data are in SI units with the length scale 1 m inside the sine ($y = \sin x$ means $y = (1\ \text{m})\sin(x/1\ \text{m})$) and $\dot{x} = 1\ \text{m/s}$. Then $[v] = \text{m/s}$, $[a_t] = [a_n] = \text{m/s}^2$ and $[\rho] = [v^2/|a_n|] = \text{m}$, and $[s] = [\int \sqrt{1+\cos^2 u}\, du] = \text{m}$.
- **Special points:** at the inflection points $x = k\pi$, $\sin x = 0$, so $a_n = 0$ and $\rho \to \infty$ (locally straight). At the crests $x = \pi/2$: $v = 1\ \text{m/s}$, $a_t = 0$, $a_n = -1\ \text{m/s}^2$ and $\rho = v^2/|a_n| = 1$ m, equal to $1/|y''| = 1$ m. At $x = \pi/4$: $\rho = (1+\cos^2 x)^{3/2}/|\sin x| = 3\sqrt{3}/2 = 2.598$ m.
- **Arc length:** $s(\pi) = \int_0^\pi \sqrt{1+\cos^2 u}\, du = 3.8202$ m, greater than the horizontal distance π m, as it must be.
- **Symbolic check (sympy):** $a_t^2 + a_n^2 = \sin^2 x = \|\mathbf{a}\|^2$ and $v^2/|a_n| = (1+\cos^2 x)^{3/2}/|\sin x|$ hold with zero residual.

Source: Problems.pdf (Page 7)

PROBLEM 07

Skater Pulled by Vehicle on a Circumference

Original statement · Problems.pdf, p. 8

Relative Kinematics & Bar Constraint

A skater B is being towed by a vehicle A on a perfectly smooth horizontal plane, where we define an inertial reference frame Oxy. The vehicle is moving along a circumference of radius a and center at O with constant angular velocity ω. The skater is attached to the vehicle by a bar without mass and constant length $l = a$. Initially, the vehicle is located at $(a, 0)$ and the skater is at rest at $(2a, 0)$. Defining a rotating reference frame Ox_1y_1 with origin at O and axis Ox_1 along OA, and calling ψ the angle between the axis Ox_1 and the bar AB, determine:

(a) The relative acceleration of the skater with respect to Ox_1y_1 as a function of ψ and its time derivatives.

(b) The centripetal and Coriolis acceleration terms also as a function of ψ and its time derivatives.

(c) Reduce the evolution of ψ to integrals of the form $\int [...] d\psi = \int [...] dt$ (consider that $\dot{\psi}^2 = \int 2\dot{\psi}\ddot{\psi}\, dt$).

(d) Determine the tension on the bar.

(e) If the rigid bar is substituted by a string which cannot bear compression, determine if/when the string will become loose.

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** A vehicle A moves counterclockwise at constant speed ω along a circle of radius a. Skater B on a frictionless horizontal plane is towed by vehicle A via a massless bar of length $l = a$.
- **Initial Conditions:** Vehicle A is at $(a, 0)$, skater B is at rest at $(2a, 0)$, meaning at $t = 0$, $\psi(0) = 0$. Because the skater is at rest in the inertial frame, $\dot{\psi}(0) = -2\omega$ (derived in part (c)).
- **Parameters & Units:** Radius and bar length a $[\mathrm{m}]$, angular speed ω $[\mathrm{rad/s}]$, skater mass m $[\mathrm{kg}]$.

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Inertial Frame:** $\mathcal{S}_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$.
- **Rotating Frame Attached to OA:** $\mathcal{S}_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$, with $\mathbf{i}_1$ pointing from O to vehicle A.

$$\boldsymbol{\omega}_{1/0} = \omega \mathbf{k}_0 = \omega \mathbf{k}_1$$

$$\mathbf{r}_A = a \mathbf{i}_1$$

- **Bar Orientation:** The bar AB is directed at angle ψ relative to $\mathbf{i}_1$:

$$\mathbf{r}_{B/A} = a \cos\psi \, \mathbf{i}_1 + a \sin\psi \, \mathbf{j}_1$$

- **Position of Skater B in Rotating Frame:**

$$\mathbf{r}_B = \mathbf{r}_A + \mathbf{r}_{B/A} = a(1+\cos\psi)\mathbf{i}_1 + a\sin\psi\,\mathbf{j}_1$$

- **Change of basis $\mathcal{B}_0 \to \mathcal{B}_1$.** The vehicle starts at $(a,0)$ and turns counterclockwise at rate ω, so $\mathbf{i}_1 = \cos\omega t\,\mathbf{i}_0 + \sin\omega t\,\mathbf{j}_0$ and the columns of the matrix below are the components of $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ in $\mathcal{B}_0$:

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & -\sin\omega t & 0 \\ \sin\omega t & \cos\omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos^2\omega t + \sin^2\omega t = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

- **Bar basis $\mathcal{B}_1 \to \mathcal{B}_{\text{bar}}$.** The pair $\mathbf{u}_{AB} = \cos\psi\,\mathbf{i}_1 + \sin\psi\,\mathbf{j}_1$, $\mathbf{u}_{\perp} = -\sin\psi\,\mathbf{i}_1 + \cos\psi\,\mathbf{j}_1$ used in parts (c) and (d) is obtained with

$$[{}_1R_{\text{bar}}] = \begin{pmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_1R_{\text{bar}}] = 1, \qquad [{}_1R_{\text{bar}}][{}_1R_{\text{bar}}]^T = I$$

- The composition $[{}_0R_{\text{bar}}] = [{}_0R_1][{}_1R_{\text{bar}}]$ is the planar rotation by $\omega t + \psi$ (the bar direction measured from the inertial x axis). The determinants and the products RR^T were verified symbolically with sympy.

PHASE 3

Step-by-Step Mathematical Deduction

(a) Relative Acceleration in $\mathcal{S}_1$

Pedagogical Justification: The relative velocity and acceleration are the time derivatives of $\mathbf{r}_B$ observed within frame $\mathcal{S}_1$, treating unit vectors $\mathbf{i}_1, \mathbf{j}_1$ as constant.

- **Relative Velocity:**

$$\mathbf{v}_{\text{rel}} = \frac{d\mathbf{r}_B}{dt}\Big|_{\mathcal{S}_1} = -a\dot{\psi}\sin\psi\,\mathbf{i}_1 + a\dot{\psi}\cos\psi\,\mathbf{j}_1$$

- **Relative Acceleration (Explicit Product Rule):**

$$\mathbf{a}_{\text{rel}} = \frac{d\mathbf{v}_{\text{rel}}}{dt}\Big|_{\mathcal{S}_1} = \frac{d}{dt}(-a\dot{\psi}\sin\psi)\mathbf{i}_1 + \frac{d}{dt}(a\dot{\psi}\cos\psi)\mathbf{j}_1$$

$$= -a(\ddot{\psi}\sin\psi + \dot{\psi}^2\cos\psi)\mathbf{i}_1 + a(\ddot{\psi}\cos\psi - \dot{\psi}^2\sin\psi)\mathbf{j}_1$$

(b) Coriolis and Centripetal Accelerations

> **Pedagogical Justification:** By Coriolis' theorem of relative kinematics (Notes Sec. 3.2, Eq. 3.27):

$$\mathbf{a}_{\text{cor}} = 2\boldsymbol{\omega}_{1/0} \times \mathbf{v}_{\text{rel}}$$

$$\mathbf{a}_{\text{cf}} = \boldsymbol{\omega}_{1/0} \times (\boldsymbol{\omega}_{1/0} \times \mathbf{r}_B) = -\omega^2 \mathbf{r}_{B\perp}$$

- **Coriolis Calculation:**

$$\mathbf{a}_{\text{cor}} = 2(\omega\mathbf{k}_1) \times (-a\dot{\psi}\sin\psi\,\mathbf{i}_1 + a\dot{\psi}\cos\psi\,\mathbf{j}_1)$$

Using $\mathbf{k}_1 \times \mathbf{i}_1 = \mathbf{j}_1$ and $\mathbf{k}_1 \times \mathbf{j}_1 = -\mathbf{i}_1$:

$$\mathbf{a}_{\text{cor}} = 2\omega \left(-a\dot{\psi} \sin\psi \, \mathbf{j}_1 - a\dot{\psi} \cos\psi \, \mathbf{i}_1 \right) = -2a\omega\dot{\psi}(\cos\psi \, \mathbf{i}_1 + \sin\psi \, \mathbf{j}_1)$$

(Compare with the official solution key (Problems.pdf, Solution keys section).)

- **Centripetal / Centrifugal Acceleration Term:**

Since $\mathbf{r}_B$ lies entirely in the x_1y_1 plane, $\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}_B) = -\omega^2 \mathbf{r}_B$:

$$\mathbf{a}_{\text{cf}} = -\omega^2 \mathbf{r}_B = -a\omega^2[(1 + \cos\psi)\mathbf{i}_1 + \sin\psi \, \mathbf{j}_1]$$

(Compare with the official solution key (Problems.pdf, Solution keys section).)

(c) Reduction of Equation of Motion to Integrals (Quadrature)

> **Pedagogical Justification:** The only force acting on skater B in the horizontal plane is the tension $\mathbf{T}$ from the bar, which acts along the bar: $\mathbf{u}_{AB} = \cos\psi \, \mathbf{i}_1 + \sin\psi \, \mathbf{j}_1$. Therefore, the total acceleration of B in the direction perpendicular to the bar $\mathbf{u}_\perp = -\sin\psi \, \mathbf{i}_1 + \cos\psi \, \mathbf{j}_1$ must vanish: $\mathbf{a}_B \cdot \mathbf{u}_\perp = 0$.

- **Projection of Acceleration:**

$$\mathbf{a}_{\text{total}} = \mathbf{a}_{\text{rel}} + \mathbf{a}_{\text{cor}} + \mathbf{a}_{\text{cf}}$$

Taking the dot product with $\mathbf{u}_\perp$:

- $\mathbf{a}_{\text{rel}} \cdot \mathbf{u}_\perp = a\ddot{\psi}(\sin^2\psi + \cos^2\psi) = a\ddot{\psi}$.
- $\mathbf{a}_{\text{cor}} \cdot \mathbf{u}_\perp = -2a\omega\dot{\psi}(\cos\psi(-\sin\psi) + \sin\psi\cos\psi) = 0$.
- $\mathbf{a}_{\text{cf}} \cdot \mathbf{u}_\perp = -a\omega^2[(1 + \cos\psi)(-\sin\psi) + \sin\psi\cos\psi] = -a\omega^2(-\sin\psi - \sin\psi\cos\psi + \sin\psi\cos\psi) = a\omega^2\sin\psi$.

Therefore:

$$a\ddot{\psi} + a\omega^2 \sin\psi = 0 \implies \ddot{\psi} + \omega^2 \sin\psi = 0$$

Multiplying by $2\dot{\psi}$ and using the chain rule $\frac{d}{dt}(\dot{\psi}^2) = 2\dot{\psi}\ddot{\psi}$ and $\frac{d}{dt}(\cos\psi) = -\sin\psi \, \dot{\psi}$:

$$2\dot\psi\ddot\psi = -2\omega^2 \sin\psi\,\dot\psi \implies \frac{d}{dt}(\dot\psi^2) = -2\omega^2\sin\psi\,\dot\psi = 2\omega^2\frac{d}{dt}(\cos\psi)$$

Initial conditions. The skater starts at rest in the inertial frame, $\mathbf{v}_B(0) = \mathbf{0}$, at $\mathbf{r}_B(0) = 2a\,\mathbf{i}_1$ (so $\psi(0) = 0$). Since $\mathbf{v}_B = \boldsymbol{\omega}_{1/0}\times\mathbf{r}_B + \mathbf{v}_{\text{rel}}$:

$$\mathbf{v}_{\text{rel}}(0) = -\omega\mathbf{k}_1 \times 2a\,\mathbf{i}_1 = -2a\omega\,\mathbf{j}_1, \qquad \mathbf{v}_{\text{rel}}(0) = a\dot\psi(0)\,\mathbf{j}_1 \implies \dot\psi(0) = -2\omega$$

Integrating with Barrow's rule between $t = 0$ ($\psi = 0$, $\dot\psi = -2\omega$) and a generic instant, with u a dummy variable ($du = \dot\psi\,dt$ along the motion):

$$\dot\psi^2 - (-2\omega)^2 = 2\omega^2\int_0^{\psi}(-\sin u)\,du = 2\omega^2\,[\cos u]_0^{\psi} = 2\omega^2(\cos\psi - 1)$$

$$\dot\psi^2 = 2\omega^2(1+\cos\psi) = 4\omega^2\cos^2\!\left(\frac{\psi}{2}\right) \quad \text{using } 1+\cos\psi = 2\cos^2(\psi/2)$$

Taking the square root, $\dot\psi = \pm 2\omega\cos(\psi/2)$. For $-\pi < \psi < \pi$ the factor $\cos(\psi/2)$ is positive, so the sign cannot change, and $\dot\psi(0) = -2\omega$ selects the minus sign (ψ decreases from zero):

$$\dot\psi = -2\omega\cos\left(\frac{\psi}{2}\right) \quad \text{for } -\pi < \psi \le 0$$

This agrees with the official key (Problems.pdf, PDF page 72).

Separating variables and integrating from $(t,\psi) = (0,0)$, with the substitution $w = u/2$, $du = 2\,dw$:

$$\int_0^{\psi} \frac{du}{2\cos(u/2)} = -\omega \int_0^{t} dt' \implies \int_0^{\psi/2} \frac{dw}{\cos w} = \ln\left|\sec\frac{\psi}{2} + \tan\frac{\psi}{2}\right| = -\omega t$$

(d) Tension on the Bar

> **Pedagogical Justification:** The tension is obtained from the radial force balance along the bar:

$$T = -m(\mathbf{a}_{\text{total}} \cdot \mathbf{u}_{AB})$$

Evaluating the terms along $\mathbf{u}_{AB} = \cos\psi\,\mathbf{i}_1 + \sin\psi\,\mathbf{j}_1$:

- $\mathbf{a}_{\text{rel}} \cdot \mathbf{u}_{AB} = -a(\ddot{\psi}\sin\psi + \dot{\psi}^2\cos\psi)\cos\psi + a(\ddot{\psi}\cos\psi - \dot{\psi}^2\sin\psi)\sin\psi = -a\dot{\psi}^2$.
- $\mathbf{a}_{\text{cor}} \cdot \mathbf{u}_{AB} = -2a\omega\dot{\psi}(\cos^2\psi + \sin^2\psi) = -2a\omega\dot{\psi}$.
- $\mathbf{a}_{\text{cf}} \cdot \mathbf{u}_{AB} = -a\omega^2[(1+\cos\psi)\cos\psi + \sin^2\psi] = -a\omega^2(1+\cos\psi)$.

$$T = ma\left[\dot{\psi}^2 + 2\omega\dot{\psi} + \omega^2(1+\cos\psi)\right]$$

With $c = \cos(\psi/2)$: $\dot{\psi}^2 = 4\omega^2 c^2$, $\dot{\psi} = -2\omega c$, $1+\cos\psi = 2c^2$, hence $T = ma\omega^2(4c^2 - 4c + 2c^2) = ma\omega^2(6c^2 - 4c)$:

$$T = 2ma\omega^2\cos\left(\frac{\psi}{2}\right)\left(3\cos\left(\frac{\psi}{2}\right) - 2\right)$$

(Compare with the official solution key (Problems.pdf, Solution keys section).)

(e) Loose String Condition

Pedagogical Justification: A flexible string cannot support compression ($T \geq 0$). The string becomes loose when the tension drops to zero: $T = 0$.

$$3\cos\left(\frac{\psi}{2}\right) - 2 = 0 \implies \cos\left(\frac{\psi}{2}\right) = \frac{2}{3}$$

$$\frac{\psi}{2} = -\arccos\left(\frac{2}{3}\right) \implies \psi = -2\arccos\left(\frac{2}{3}\right) \approx -96.38^\circ$$

(Compare with the official solution key (Problems.pdf, Solution keys section).)

PHASE 4

Physical Interpretation, Limiting Cases and Dimensional Check

- **Dimensional check:**
- $[\ddot{\psi}] = [\omega^2 \sin\psi] = \mathrm{s}^{-2}$ and $[\dot{\psi}^2] = [2\omega^2(1+\cos\psi)] = \mathrm{s}^{-2}$ (the angle is dimensionless).
- $[T] = [m\,a\,\omega^2] = \mathrm{kg}\cdot\mathrm{m}\cdot\mathrm{s}^{-2} = \mathrm{N}$.
- **Limit $\omega \to 0$:** the vehicle is at rest, the skater is at rest, and $T \to 0$, $\dot{\psi} \to 0$, as expected.
- **Independent check of T at $t = 0$ ($\psi = 0$).** The closed form gives $T(0) = 2ma\omega^2 \cdot 1 \cdot (3-2) = 2ma\omega^2$. For a rigid bar of length $l = a$, the relative velocity $\mathbf{v}_B - \mathbf{v}_A$ is perpendicular to the bar, hence $(\mathbf{a}_B - \mathbf{a}_A)\cdot\mathbf{u}_{AB} = -\|\mathbf{v}_B - \mathbf{v}_A\|^2/a$. At $t = 0$: $\mathbf{v}_B = \mathbf{0}$, $\|\mathbf{v}_A\| = a\omega$, $\mathbf{a}_A = -a\omega^2\mathbf{i}_0$ and $\mathbf{u}_{AB} = \mathbf{i}_0$, so $\mathbf{a}_B\cdot\mathbf{u}_{AB} = -a\omega^2 - a\omega^2 = -2a\omega^2$ and $T = 2ma\omega^2$ (Consistent).
- **Symbolic check (sympy).** Differentiating twice, in the inertial frame, the position $\mathbf{r}_B = [{}_0R_1]\,a\,(1+\cos\psi,\ \sin\psi,\ 0)^T$ and projecting on $\mathbf{u}_{AB}$ and $\mathbf{u}_\perp$ gives $\mathbf{a}_B\cdot\mathbf{u}_\perp = a(\ddot{\psi} + \omega^2\sin\psi)$ and $T = ma\left[\dot{\psi}^2 + 2\omega\dot{\psi} + \omega^2(1+\cos\psi)\right]$, identical to parts (c) and (d); substituting $\dot{\psi} = -2\omega\cos(\psi/2)$ reproduces $T = 2ma\omega^2\cos(\psi/2)\,(3\cos(\psi/2) - 2)$ with zero residual.
- **When the string goes loose (numerical).** The closed form $\omega t = -\ln|\sec(\psi/2) + \tan(\psi/2)|$ evaluated at $\cos(\psi/2) = 2/3$ gives $\omega t_{\text{loose}} = \ln\dfrac{2}{3-\sqrt{5}} = \ln\dfrac{3+\sqrt{5}}{2} = 0.96242$. An RK4 integration (step 10^{-5}, $\omega = 1$) of $\ddot{\psi} = -\omega^2\sin\psi$ with

$\psi(0)=0, \dot\psi(0)=-2\omega$ reaches $\cos(\psi/2)=2/3$ at $\omega t = 0.96242$ with $\psi=-96.379^\circ$, in agreement with $-2\arccos(2/3)=-96.379^\circ$; there $\dot\psi^2 = 1.777787\,\omega^2$ against $2\omega^2(1+\cos\psi)=1.777787\,\omega^2$.

Source: Problems.pdf (Page 8)

PROBLEM 08

Kinematics, Two Trains on Tracks

Original statement · Problems.pdf, p. 8

Relative Motion

Two trains, A and B, are moving along parallel tracks. The separation between the tracks is a. The absolute velocity of each train is v_A and v_B, respectively. What is the relative velocity of each train with respect to the other?

Solve the same problem considering that train B is now moving along a circular track of radius R. The minimum distance between the circular track and the straight track of A is a.

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** Two trains A and B move on separate tracks with distance of closest separation a.
- Case 1: Two parallel straight tracks with constant speeds v_A and v_B.
- Case 2: Train A on a straight track, Train B on a circular track of radius R.
- **Parameters & Units:** Separation a $[\mathrm{m}]$, radius R $[\mathrm{m}]$, speeds v_A, v_B $[\mathrm{m/s}]$.

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Inertial Cartesian Frame:** $\mathcal{S}_0 = \{O; \mathbf{i}_0, \mathbf{j}_0\}$.

PHASE 3

Step-by-Step Mathematical Deduction

Case 1: Parallel Tracks

- Train A on line $y = 0$: $\mathbf{r}_A(t) = v_A t\,\mathbf{i}_0$.
- Train B on line $y = a$: $\mathbf{r}_B(t) = (x_{B0} + v_B t)\mathbf{i}_0 + a\mathbf{j}_0$.
- **Relative Velocity:**

$$\mathbf{v}_{B/A} = \mathbf{v}_B - \mathbf{v}_A = (v_B - v_A)\mathbf{i}_0$$

$$\mathbf{v}_{A/B} = -\mathbf{v}_{B/A} = (v_A - v_B)\mathbf{i}_0$$

Case 2: Train B on a Circular Track of Radius R

- Straight track of A along the x-axis: $\mathbf{r}_A = v_A t\,\mathbf{i}_0$.
- Circular track center at $(0, R + a)$ so that the closest distance to A's track is $(R + a) - R = a$.
- Position of train B traveling at angular speed $\omega_B = \frac{v_B}{R}$ with angle $\theta_B(t) = \omega_B t$:

$$\mathbf{r}_B(t) = R\sin(\omega_B t)\mathbf{i}_0 + [R + a - R\cos(\omega_B t)]\mathbf{j}_0$$

- Velocity of train B:

$$\mathbf{v}_B = \frac{d\mathbf{r}_B}{dt} = R\omega_B\cos(\omega_B t)\mathbf{i}_0 + R\omega_B\sin(\omega_B t)\mathbf{j}_0 = v_B\cos\left(\frac{v_B t}{R}\right)\mathbf{i}_0 + v_B\sin\left(\frac{v_B t}{R}\right)\mathbf{j}_0$$

- **Relative Velocity $\mathbf{v}_{B/A}$:**

$$\mathbf{v}_{B/A} = \mathbf{v}_B - \mathbf{v}_A = \left[v_B \cos\left(\frac{v_B t}{R}\right) - v_A\right]\mathbf{i}_0 + v_B \sin\left(\frac{v_B t}{R}\right)\mathbf{j}_0$$

- **Magnitude of Relative Velocity:**

$$v_{\text{rel}} = \sqrt{\left[v_B \cos\left(\frac{v_B t}{R}\right) - v_A\right]^2 + v_B^2 \sin^2\left(\frac{v_B t}{R}\right)} = \sqrt{v_B^2 + v_A^2 - 2v_A v_B \cos\left(\frac{v_B t}{R}\right)}$$

PHASE 4

Physical Interpretation, Limiting Cases and Dimensional Check

- **Dimensional check:** $[\mathbf{v}_{B/A}] = \mathrm{m/s}$; in Case 2 the argument of the trigonometric functions is $\dfrac{v_B t}{R} = \dfrac{(\mathrm{m/s})\,\mathrm{s}}{\mathrm{m}}$, which is dimensionless (rad), and $[v_B^2 + v_A^2 - 2v_A v_B \cos(\cdot)] = \mathrm{m}^2/\mathrm{s}^2$, so v_{rel} is in m/s.
- **Role of the separation a:** it does not appear in any relative velocity, because $\mathbf{v}_{B/A}$ is the time derivative of $\mathbf{r}_B - \mathbf{r}_A$ and a only enters the constant offset of that vector.
- **Limiting cases of Case 2 (sympy):**
- $t = 0$: $v_{\text{rel}} = \sqrt{v_A^2 + v_B^2 - 2v_A v_B} = |v_B - v_A|$, because both velocities are parallel to $\mathbf{i}_0$ at that instant.
- $R \to \infty$ (the circular track straightens): $\cos(v_B t/R) \to 1$ and $v_{\text{rel}} \to |v_B - v_A|$, which is the Case 1 result.
- $v_A = v_B = v$: $v_{\text{rel}} = v\sqrt{2 - 2\cos(vt/R)} = 2v\left|\sin\dfrac{vt}{2R}\right|$, which starts at 0 and never exceeds $2v$.
- $v_A = 0$: $v_{\text{rel}} = v_B$ for all t (relative to a stationary train, only the speed of B matters).
- **Symbolic check:** $\|\mathbf{v}_B - \mathbf{v}_A\|^2$ computed from the components equals $v_B^2 + v_A^2 - 2v_A v_B \cos(v_B t/R)$ with zero residual. Both cases are written in the single inertial basis $\mathcal{B}_0$, so no change-of-basis matrix is needed.

Source: Problems.pdf (Page 8)

PROBLEM 09

2001: A Space Odyssey (Artificial Gravity)

Original statement · Problems.pdf, p. 8

Normal Acceleration & Rotating Cylinder

In *2001: A Space Odyssey* an idea to obtain artificial gravity in a spaceship was presented. It consists of a cylinder rotating at constant angular velocity around its axis. The astronauts would live in the inner surface of the cylinder and the centrifugal force would act as artificial gravity.

(a) If the cylinder has radius R, determine the angular velocity ω required to obtain a given level of artificial gravity g. Consider $R = 10\ \text{m}$ and $g = 9.81\ \text{m/s}^2$ for a particular case.

(b) An astronaut runs inside the surface of the cylinder with a constant speed u with respect to the cylinder. Determine the artificial gravity that she feels (neglect the height of the astronaut h with respect to the radius of the cylinder R, $h \ll R$).

(c) What is the condition for her to feel no artificial gravity at all?

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** A rotating cylindrical spacecraft hull of radius R generates artificial gravity via centrifugal normal acceleration.
- **Hypotheses:** Spacecraft rotates at constant rate ω about its symmetry axis in deep space (no external gravitational field).
- **Parameters & Units:** Radius $R = 10\ \text{m}$, target gravity $g = 9.81\ \text{m/s}^2$, relative running speed u $[\text{m/s}]$.

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- Intrinsic acceleration of a point on the rim rotating at speed $v_{\text{rim}} = \omega R$:

$$a_n = \frac{v_{\text{rim}}^2}{R} = \omega^2 R$$

- **Hull-fixed rotating basis.** Take the inertial basis $\mathcal{B}_0 = \{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ with $\mathbf{k}_0$ along the axis of the cylinder, and the basis $\mathcal{B}_1 = \{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}_0\}$ attached to the hull, rotating at constant rate ω about $\mathbf{k}_0$ (the angle is $\theta = \omega t$ for a point fixed to the hull). The columns are the components of $\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}_0$ in $\mathcal{B}_0$:

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & -\sin\omega t & 0 \\ \sin\omega t & \cos\omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos^2\omega t + \sin^2\omega t = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

- By Poisson's formula with $\boldsymbol{\omega} = \omega \mathbf{k}_0$: $\dot{\mathbf{e}}_r = \omega \mathbf{e}_\theta$ and $\dot{\mathbf{e}}_\theta = -\omega \mathbf{e}_r$. A point of the rim is $\mathbf{r} = R\,\mathbf{e}_r$, hence $\mathbf{v} = \omega R\,\mathbf{e}_\theta$ and $\mathbf{a} = -\omega^2 R\,\mathbf{e}_r$, which is the intrinsic acceleration $a_n = \omega^2 R$ above, directed towards the axis. An astronaut running at speed u relative to the floor along $\pm\mathbf{e}_\theta$ has absolute velocity $(\omega R \pm u)\,\mathbf{e}_\theta$. The determinant and the product RR^T were verified symbolically with sympy.

PHASE 3

Step-by-Step Mathematical Deduction

(a) Angular Velocity for Target Artificial Gravity g

Pedagogical Justification: The apparent weight felt by an astronaut standing still on the inner hull is the normal reaction provided by the floor: $N = m a_n = m\omega^2 R$. For this reaction to match standard Earth gravity mg:

$$\omega^2 R = g \implies \omega = \sqrt{\frac{g}{R}}$$

- **Numerical Calculation for $R = 10\ \text{m}, g = 9.81\ \text{m/s}^2$:**

$$\omega = \sqrt{\frac{9.81\ \text{m/s}^2}{10\ \text{m}}} = \sqrt{0.981}\ \text{s}^{-1} \approx 0.99045\ \text{rad/s}$$

In revolutions per minute (RPM):

$$N_{\text{rpm}} = \omega \times \frac{60}{2\pi} = 0.99045 \times 9.5493 \approx 9.46\ \text{rpm}$$

(b) Artificial Gravity Felt by a Running Astronaut

> **Pedagogical Justification:** When the astronaut runs along the circumference at speed u relative to the floor, her absolute speed in the inertial frame is $v_{\text{abs}} = \omega R \pm u$, where $(+)$ corresponds to running in the direction of rotation (prograde/eastward) and $(-)$ corresponds to running against rotation (retrograde/westward).

- **Apparent Gravity:**

$$g_{\text{felt}} = a_n = \frac{v_{\text{abs}}^2}{R} = \frac{(\omega R \pm u)^2}{R} = \omega^2 R \left(1 \pm \frac{u}{\omega R}\right)^2 = g\left(1 \pm \frac{u}{\omega R}\right)^2$$

Expanding:

$$g_{\text{felt}} = \frac{\omega^2 R^2 \pm 2\omega R u + u^2}{R} = g \pm 2\omega u + \frac{u^2}{R}$$

(The term $2\omega u$ represents the Coriolis acceleration felt by the running astronaut!)

(c) Condition to Feel No Artificial Gravity At All

Pedagogical Justification: For the astronaut to become completely weightless ($g_{\text{felt}} = 0$):

$$\frac{(\omega R \pm u)^2}{R} = 0 \implies \omega R \pm u = 0$$

This requires choosing the retrograde direction $(-)$ and matching the speed:

$$u = \omega R$$

- **Physical Explanation:** Running backwards at the exact hull speed cancels out her inertial circular motion. She becomes stationary relative to the non-rotating inertial space, floating weightlessly while the floor slips beneath her feet!

PHASE 4

Physical Interpretation, Limiting Cases and Dimensional Check

- **Dimensional check:** $[\omega] = \sqrt{[g]/[R]} = \sqrt{(\mathrm{m/s^2})/\mathrm{m}} = \mathrm{s^{-1}}$; $[\omega^2 R] = \mathrm{m/s^2}$; $[2\omega u] = \mathrm{s^{-1}} \cdot \mathrm{m/s} = \mathrm{m/s^2}$; $[u^2/R] = (\mathrm{m^2/s^2})/\mathrm{m} = \mathrm{m/s^2}$ (Consistent).
- **Numerical values (Python, $R = 10$ m, $g = 9.81$ m/s^2):**
- $\omega = \sqrt{0.981} = 0.990454$ rad/s $= 9.458$ rpm (quoted above as ≈ 9.46 rpm) and rim speed $\omega R = 9.9045$ m/s.
- Running at $u = 2$ m/s: prograde $g_{\text{felt}} = 14.17\ \mathrm{m/s^2}$ ($1.44\,g$), retrograde $6.25\ \mathrm{m/s^2}$ ($0.64\,g$). At $u = 5$ m/s: prograde $22.21\ \mathrm{m/s^2}$ ($2.26\,g$), retrograde $2.41\ \mathrm{m/s^2}$ ($0.25\,g$).
- Retrograde at $u = \omega R = 9.9045$ m/s gives $g_{\text{felt}} = 0$ exactly (part (c)); prograde at the same speed gives $4g = 39.24\ \mathrm{m/s^2}$.
- **Limiting cases:** $u \to 0$ gives $g_{\text{felt}} = g$. For $u \ll \omega R$, $g_{\text{felt}} \approx g \pm 2\omega u$, the Coriolis correction (at $u = 2$ m/s: $9.81 + 3.96 + 0.40 = 14.17\ \mathrm{m/s^2}$, where $0.40 = u^2/R$ is the remaining quadratic term, so the expansion of part (b) is reproduced exactly). For $R \to \infty$ at fixed g, $\omega = \sqrt{g/R} \to 0$ (a larger hull needs fewer revolutions per minute).

Source: Problems.pdf (Page 8)

PROBLEM 41

Simple Rotations, Satellite Solar Panels

Original statement · Problems.pdf, p. 29

Rotational Kinematics

The angle θ describing the rotation of the solar panels of a reconnaissance satellite about the body-fixed axis x is an arbitrary function of time. The satellite spins about the z axis at the constant angular rate ω. Derive expressions for the absolute velocity and acceleration of point B in the figure relative to an inertial reference frame centered in A.

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** A reconnaissance satellite spins about its body-fixed z-axis at constant rate ω. A solar panel rotates about the body-fixed x-axis by an arbitrary prescribed angle $\theta(t)$.
- **Goal:** Determine absolute velocity $\mathbf{v}_{B/0}$ and acceleration $\mathbf{a}_{B/0}$ of tip B relative to inertial frame centered at A.
- **Geometric Parameters:** H is the distance from A to B measured along the x-axis and W is the distance from the x-axis to B (panel width), as read from the figure in Problems.pdf (PDF page 29).

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Inertial Reference Frame:** $\mathcal{S}_0 = \{A; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$.
- **Satellite Body Frame (Precessing):** $\mathcal{S}_1 = \{A; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$, obtained by rotating by angle $\phi = \omega t$ about $\mathbf{k}_0$:

$$[{}_0R_1] = \begin{pmatrix} \cos(\omega t) & -\sin(\omega t) & 0 \\ \sin(\omega t) & \cos(\omega t) & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \boldsymbol{\omega}_{1/0} = \omega \mathbf{k}_1$$

- **Panel-Fixed Frame:** $\mathcal{S}_2 = \{C; \mathbf{i}_2, \mathbf{j}_2, \mathbf{k}_2\}$, obtained by rotating $\mathcal{S}_1$ by angle $\theta(t)$ about $\mathbf{i}_1$:

$$[{}_1R_2] = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix}, \quad \boldsymbol{\omega}_{2/1} = \dot{\theta}\mathbf{i}_1$$

- **Total Angular Velocity Vector (Notes Sec. 3.1, Eq. 3.14):**

$$\boldsymbol{\omega}_{2/0} = \boldsymbol{\omega}_{1/0} + \boldsymbol{\omega}_{2/1} = \dot{\theta}\mathbf{i}_1 + \omega\mathbf{k}_1$$

- **Angular Acceleration Vector:**

$$\boldsymbol{\alpha}_{2/0} = \left.\frac{d\boldsymbol{\omega}_{2/0}}{dt}\right|_{\mathcal{S}_0} = \ddot{\theta}\mathbf{i}_1 + \dot{\theta}(\boldsymbol{\omega}_{1/0} \times \mathbf{i}_1) + \dot{\omega}\mathbf{k}_1 + \omega(\boldsymbol{\omega}_{1/0} \times \mathbf{k}_1)$$

Since $\omega = \text{const}$, $\dot{\omega} = 0$ and $\boldsymbol{\omega}_{1/0} \times \mathbf{k}_1 = \mathbf{0}$.

$$\boldsymbol{\alpha}_{2/0} = \ddot{\theta}\mathbf{i}_1 + \dot{\theta}(\omega\mathbf{k}_1 \times \mathbf{i}_1) = \ddot{\theta}\mathbf{i}_1 + \omega\dot{\theta}\mathbf{j}_1$$

The hinge axis is the body-fixed x-axis ($\mathbf{i}_1$); C denotes the point of this axis at the abscissa of B, so $\mathbf{r}_{AC} = H\mathbf{i}_1$ is fixed in $\mathcal{S}_1$ and does not rotate with the panel. The axes of $\mathcal{S}_2$ expressed in $\mathcal{S}_1$ are the columns of $[{}_1R_2]$: $\mathbf{i}_2 = \mathbf{i}_1$, $\mathbf{j}_2 = \cos\theta\,\mathbf{j}_1 + \sin\theta\,\mathbf{k}_1$, $\mathbf{k}_2 = -\sin\theta\,\mathbf{j}_1 + \cos\theta\,\mathbf{k}_1$. Orthonormality check: $\det[{}_1R_2] = \cos^2\theta + \sin^2\theta = 1$ and $[{}_1R_2][{}_1R_2]^T = \operatorname{diag}\left(1,\ \begin{pmatrix}\cos\theta & -\sin\theta \\ \sin\theta & \cos\theta\end{pmatrix}\begin{pmatrix}\cos\theta & \sin\theta \\ -\sin\theta & \cos\theta\end{pmatrix}\right) = I$.

Geometry assumption. The statement text does not fix the geometry; it is read from the figure in Problems.pdf (PDF page 29): the panel lies along the x-axis, B is the panel corner at distance H from A measured along x and at distance W from the x-axis, and θ is measured about $+\mathbf{i}_1$ from the $\mathbf{j}_1$ direction (so $\mathbf{r}_{CB} = W\mathbf{j}_2$). The zero reference of θ (panel along $+\mathbf{j}_1$ at $\theta = 0$) cannot be read unambiguously from the statement; it is the choice consistent with the official key.

- **Orthonormality of $[{}_0R_1]$:** $\det[{}_0R_1] = \cos^2\omega t + \sin^2\omega t = 1$ and $[{}_0R_1][{}_0R_1]^T = I$. Since $[{}_1R_2]$ has the same two properties, the composition $[{}_0R_2] = [{}_0R_1][{}_1R_2]$ is also a proper rotation ($\det = 1$, $RR^T = I$); all three checks were done symbolically with sympy.

PHASE 3

Step-by-Step Mathematical Deduction

Position of Point B in Basis $\mathcal{B}_1$

Decompose $\mathbf{r}_{AB} = \mathbf{r}_{AC} + \mathbf{r}_{CB}$. The first part is fixed in $\mathcal{S}_1$; the second is fixed in the panel frame $\mathcal{S}_2$ ($\mathbf{r}_{CB} = W\mathbf{j}_2$), and is brought to $\mathcal{S}_1$ with $[{}_1R_2]$:

$$\mathbf{r}_{AC} = H\mathbf{i}_1, \qquad \{\mathbf{r}_{CB}\}_1 = [{}_1R_2]\begin{pmatrix}0\\W\\0\end{pmatrix} = \begin{pmatrix}0\\W\cos\theta\\W\sin\theta\end{pmatrix}$$

$$\mathbf{r}_{AB} = H\mathbf{i}_1 + W\cos\theta\,\mathbf{j}_1 + W\sin\theta\,\mathbf{k}_1$$

Absolute Velocity of Point B

Since A is the origin of the inertial frame, $\mathbf{v}_{B/0} = \left.\frac{d\mathbf{r}_{AB}}{dt}\right|_{\mathcal{S}_0}$. By the Poisson formula, a vector fixed in $\mathcal{S}_1$ obeys $\left.\frac{d}{dt}\right|_{\mathcal{S}_0} = \boldsymbol{\omega}_{1/0}\times$ and a vector fixed in $\mathcal{S}_2$ obeys $\left.\frac{d}{dt}\right|_{\mathcal{S}_0} = \boldsymbol{\omega}_{2/0}\times$. The hub offset $H\mathbf{i}_1$ rotates only with the satellite ($\boldsymbol{\omega}_{1/0}$), not with the panel:

$$\mathbf{v}_{B/0} = \boldsymbol{\omega}_{1/0}\times\mathbf{r}_{AC} + \boldsymbol{\omega}_{2/0}\times\mathbf{r}_{CB} = \omega\mathbf{k}_1\times H\mathbf{i}_1 + (\dot\theta\mathbf{i}_1 + \omega\mathbf{k}_1)\times W(\cos\theta\,\mathbf{j}_1 + \sin\theta\,\mathbf{k}_1)$$

Cross products: $\mathbf{k}_1\times\mathbf{i}_1 = \mathbf{j}_1, \mathbf{i}_1\times\mathbf{j}_1 = \mathbf{k}_1, \mathbf{i}_1\times\mathbf{k}_1 = -\mathbf{j}_1, \mathbf{k}_1\times\mathbf{j}_1 = -\mathbf{i}_1, \mathbf{k}_1\times\mathbf{k}_1 = \mathbf{0}$. Term by term:

- $\omega\mathbf{k}_1 \times H\mathbf{i}_1 = H\omega\,\mathbf{j}_1$
- $\dot\theta\mathbf{i}_1 \times W\cos\theta\,\mathbf{j}_1 = W\dot\theta\cos\theta\,\mathbf{k}_1$
- $\dot\theta\mathbf{i}_1 \times W\sin\theta\,\mathbf{k}_1 = -W\dot\theta\sin\theta\,\mathbf{j}_1$
- $\omega\mathbf{k}_1 \times W\cos\theta\,\mathbf{j}_1 = -W\omega\cos\theta\,\mathbf{i}_1$
- $\omega\mathbf{k}_1 \times W\sin\theta\,\mathbf{k}_1 = \mathbf{0}$

Adding the five terms:

$$\mathbf{v}_{B/0} = -W\omega\cos\theta\,\mathbf{i}_1 + (H\omega - W\dot\theta\sin\theta)\,\mathbf{j}_1 + W\dot\theta\cos\theta\,\mathbf{k}_1$$

This agrees with the official key (Problems.pdf, PDF page 76).

Absolute Acceleration of Point B

Differentiate $\mathbf{v}_{B/0} = v_x\mathbf{i}_1 + v_y\mathbf{j}_1 + v_z\mathbf{k}_1$ in $\mathcal{S}_0$: the components are differentiated with the chain rule and product rule, and the Poisson formula gives $\dot{\mathbf{i}}_1 = \omega\mathbf{j}_1, \dot{\mathbf{j}}_1 = -\omega\mathbf{i}_1, \dot{\mathbf{k}}_1 = \mathbf{0}$:

$$\mathbf{a}_{B/0} = \dot v_x\mathbf{i}_1 + \dot v_y\mathbf{j}_1 + \dot v_z\mathbf{k}_1 + \omega\mathbf{k}_1 \times \mathbf{v}_{B/0} = (\dot v_x - \omega v_y)\mathbf{i}_1 + (\dot v_y + \omega v_x)\mathbf{j}_1 + \dot v_z\mathbf{k}_1$$

With $\frac{d}{dt}\cos\theta = -\dot\theta\sin\theta$ and $\frac{d}{dt}\sin\theta = \dot\theta\cos\theta$ (ω, H, W constant):

- $\dot v_x = W\omega\dot\theta\sin\theta, \quad v_y = H\omega - W\dot\theta\sin\theta, \quad \dot v_x - \omega v_y = 2W\omega\dot\theta\sin\theta - H\omega^2$
- $\dot v_y = -W(\ddot\theta\sin\theta + \dot\theta^2\cos\theta), \quad \omega v_x = -W\omega^2\cos\theta$
- $\dot v_z = W(\ddot\theta\cos\theta - \dot\theta^2\sin\theta)$

$$\mathbf{a}_{B/0} = (2W\omega\dot\theta\sin\theta - H\omega^2)\mathbf{i}_1 - (W\ddot\theta\sin\theta + W\dot\theta^2\cos\theta + W\omega^2\cos\theta)\mathbf{j}_1 + (W\ddot\theta\cos\theta - W\dot\theta^2\sin\theta)\mathbf{k}_1$$

This agrees with the official key (Problems.pdf, PDF page 76).

PHASE 4

Physical Interpretation & Dimensional Verification

- **Coriolis Coupling:** The term $2W\omega\dot{\theta}\sin\theta\,\mathbf{i}_1$ represents the Coriolis acceleration arising from the cross-coupling between satellite spin ω and panel deployment rate $\dot{\theta}$.
- **Dimensional Consistency:**
- Every velocity component has units $[\text{length}]\cdot[\text{time}]^{-1} = \text{m/s}$.
- Every acceleration component has units $[\text{length}]\cdot[\text{time}]^{-2} = \text{m/s}^2$ (Consistent).
- **Limiting cases:**
- $\dot{\theta} = 0$ (panel locked): $\mathbf{v}_{B/0} = -W\omega\cos\theta\,\mathbf{i}_1 + H\omega\,\mathbf{j}_1 = \omega\mathbf{k}_1 \times (H\mathbf{i}_1 + W\cos\theta\,\mathbf{j}_1 + W\sin\theta\,\mathbf{k}_1)$, the rigid-body velocity of a point of the spinning satellite.
- $\omega = 0$ (no spin): $\mathbf{v}_{B/0} = W\dot{\theta}(-\sin\theta\,\mathbf{j}_1 + \cos\theta\,\mathbf{k}_1)$ and the acceleration reduces to the circular-motion terms $-W\dot{\theta}^2\,\mathbf{e}_{\text{rad}} + W\ddot{\theta}\,\mathbf{e}_{\text{tan}}$ about the hinge axis ($\mathbf{e}_{\text{rad}} = \mathbf{j}_2$, $\mathbf{e}_{\text{tan}} = \mathbf{k}_2$, radius W).
- **Symbolic check (sympy):** differentiating twice in the inertial frame the vector $[{}_0R_1]\,(H,\ W\cos\theta,\ W\sin\theta)^T$ and projecting back on $\mathcal{B}_1$ with $[{}_0R_1]^T$ gives exactly the $\mathbf{v}_{B/0}$ and $\mathbf{a}_{B/0}$ of Phase 3 (all components, zero residual).

Source: Problems.pdf (Ginsberg, Cambridge Univ. Press, 2008)